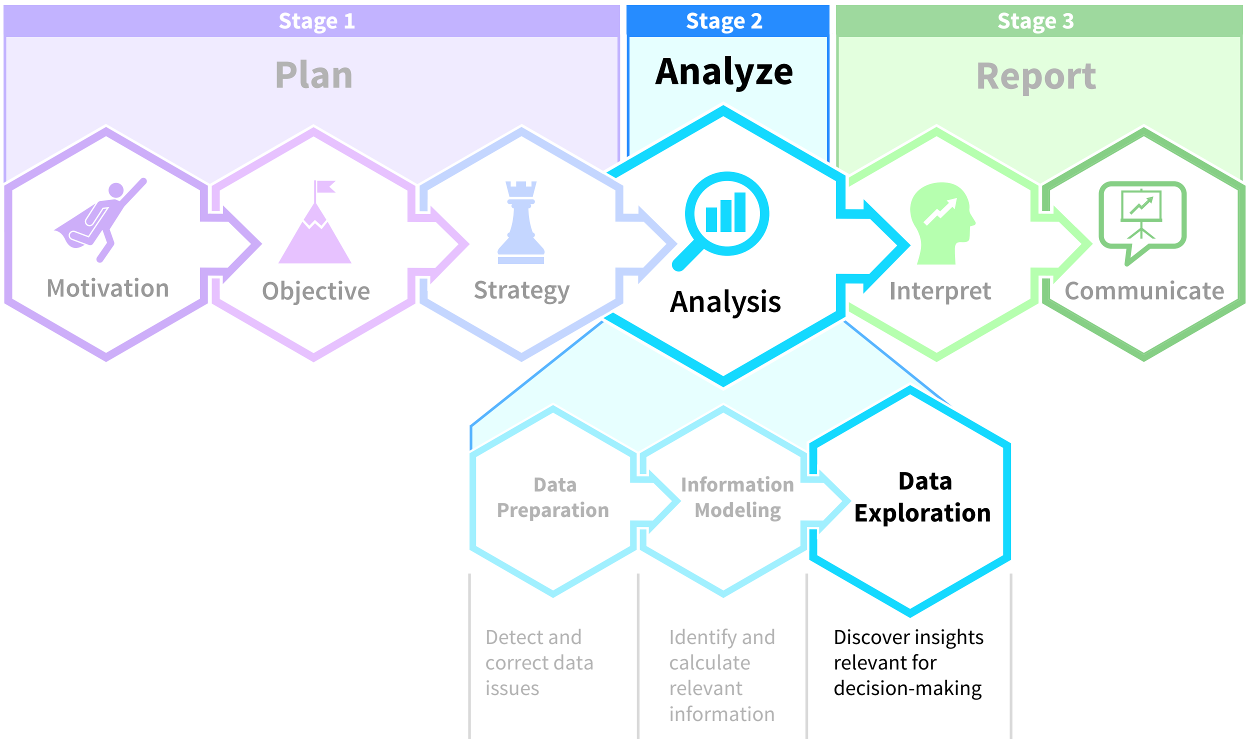
CHAPTER 7

Analysis: Data Exploration

Chapter Preview

So far in the analyze stage of the data analysis process, you have prepared the data, completed information modeling, and created an analytical database. Now, it is time to explore the data and collect insights. This step investigates data to uncover anomalies, new patterns, and anything that might improve decision-making. This chapter will help you develop the skills needed for successful data exploration, which is a process that includes asking the right questions and identifying and exploring data relationships to find insights. The patterns presented in this chapter will help you effectively and efficiently explore the data sets you will work with during your career.



Professional Insight: Why Should You Know How to Explore Data?

After earning her master's in accounting, Yuqi joined the data analytics department of a large financial institution.

> My transition into the workplace was not without challenges. **In school I was given complex problems for which I had to find elegant solutions, and I really excelled at that. However, when I started my job, there were no neatly defined questions**. For example, the financial data team continuously asked us for new insights. What can you tell us that we don't know yet? **So, I needed to learn more skills—specifically data exploration. I started looking at data sets from as many different angles as possible.** I looked at trends, distributions, grouped data in ways that no one had done before, correlated all kinds of variables, looked for outliers, and so much more. I developed a unique data exploration skill set and became good at it. **Honestly, I feel like I have become a data researcher.** What is next for me? After doing this for almost ten years, I plan to get my PhD in data science.

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 7.1 Describe the process of data exploration.	• The Process of Data Exploration • Exploring Data with PivotTables • Data Exploration Across Tools	**Explore Data with PivotTables** (Example: Managerial Accounting)
LO 7.2 Explore foundational data relationships through visualizations.	• Eight Patterns Exploring Foundational Data Relationships	**Visualize Part-To-Whole Relationships with Excel** (Example: Managerial Accounting)
LO 7.3 Explore data by integrating foundational relationships.	• Two Patterns Using a Single Visualization • Reports Using Multiple Visualizations	**Build an Interactive Report** (Example: Managerial Accounting)

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

7.1 What is Data Exploration?

LEARNING OBJECTIVE 1

Describe the process of data exploration.

In the chapter on information modeling, you learned how to add information useful for analysis to an analytical database. Now you are ready to start exploring the data. **Data exploration**, or exploratory data analysis, is the discovery process of looking for something new and previously unknown. This is accomplished by looking for patterns, outliers, or, more generally, for insights. An **insight** is an observation that might significantly affect a business' decision-making. Remember, decisions are not based on data. Rather, decisions are informed by the insights generated from data.

This process of generating insights is what distinguishes data analysis from the simple act of reporting numbers. Calculating and presenting the total dollar amount of donations per year for a non-for-profit organization is reporting, but recognizing there is a downward trend of donations over the years and that donors in different age groups behave differently is a result of data exploration. While they might share some tools, it is essential to differentiate between data exploration, interpretation, and reporting:

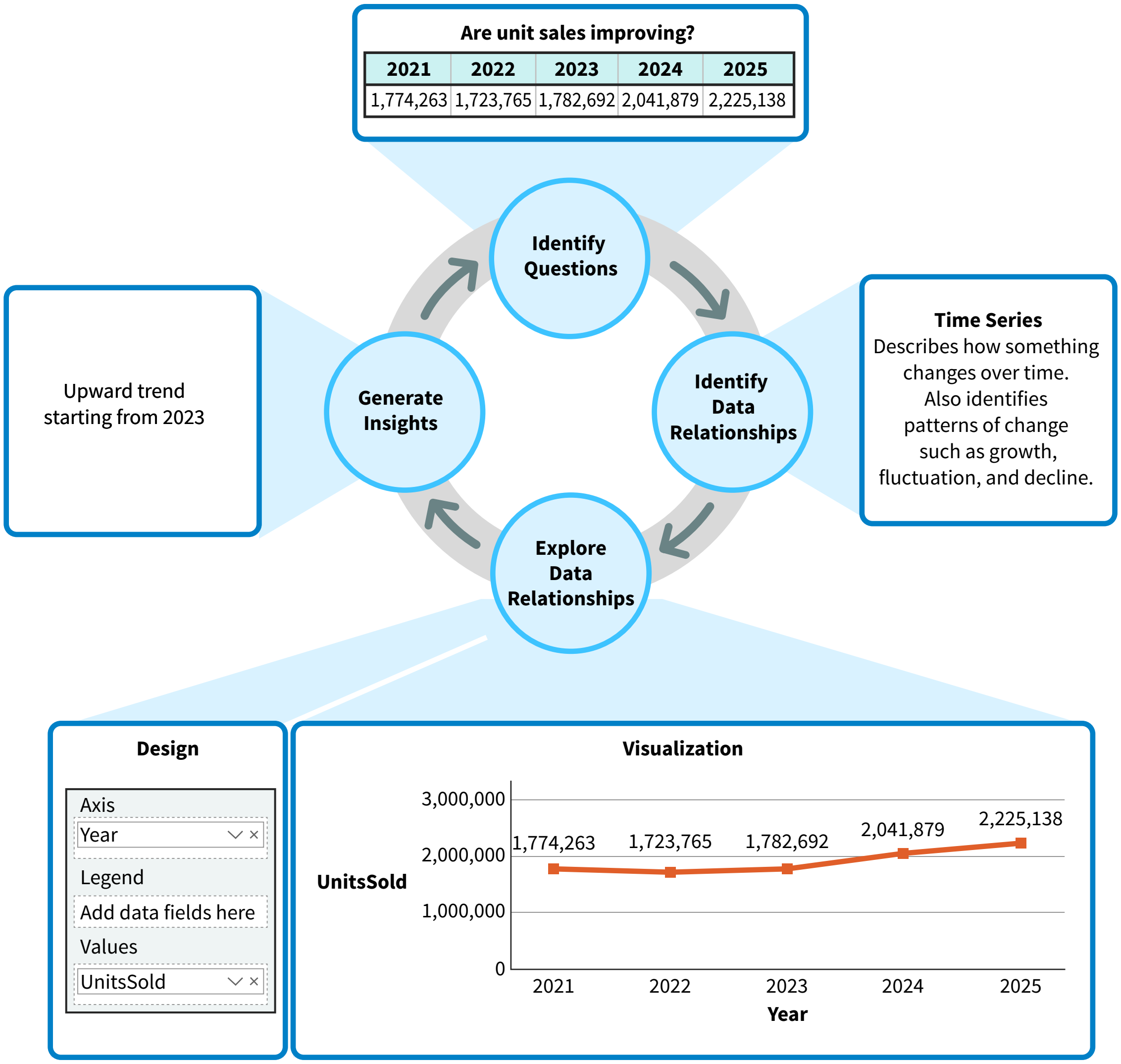
- Exploration: Discovering insights.
- Interpretation: Contextualizing and understanding insights.
- Reporting: Communicating insights.

Interpretation and reporting are covered in their own chapters. For now, let's summarize the process of data exploration.

The Process of Data Exploration

Data exploration is an integral part of what accountants do every day. They look for insights, from identifying if there are seasonal fluctuations in sales and whether those fluctuations affect the bottom line, to how changes in credit policy affect sales and uncollectible accounts receivable. It is a four-step process of identifying questions, identifying data relationships, exploring data relationships, and finally, generating insights (**Illustration 7.1**).

ILLUSTRATION 7.1 Data Exploration as a Process



Identify Questions

Data exploration helps answer accounting questions such as whether sales and profits are improving, which products deserve investment, if bad debts are appropriately managed, and more. An analytical database should provide answers to both anticipated and unanticipated (unplanned) questions. Once the question is determined, such as the question about whether unit sales are improving in Illustration 7.1, then the underlying data relationships can be identified.

Identify Data Relationships

If we want to know if sales are improving and the available data indicated a change in both the marketing budget and in unit sales, then the relationship between those two elements could generate valuable insights. A **data relationship** describes how data elements (or values) relate to each other. But before aspects of data relationships can be analyzed, they have to be identified. Stephen Few[1], an expert in data visualizations, differentiates eight foundational data relationships:

- Nominal comparison
- Distribution
- Deviation
- Ranking
- Part-to-whole
- Correlation
- Time series
- Geospatial

The data relationship identified in Illustration 7.1 is a time series, which describes how something changes over time and helps to identify patterns of change. A relationship that has been identified, whether it is a time series relationship or one of the relationships examined later in this chapter, is ready for exploration.

Explore Data Relationships

While there are different approaches to exploring data relationships, visualization and statistics are the most common. Exploration involves selecting the visualization or visualizations best suited for exploring the data relationships. In Illustration 7.1 a line chart visualizes the time series.

Keep in mind that tool-specific knowledge is required to create visualizations. For example, a time series analysis requires knowing how to create line charts. Business intelligence software such as Excel, Power BI, and Tableau all have powerful tools that visualize data relationships for exploration. The line chart in Illustration 7.1 was designed using Power BI. In the design section of the illustration, the axis field refers to the time unit used for the x-axis in the chart, and the values field refers to the variable used for the y-axis.

Generate Insights

The line chart in Illustration 7.1 shows an upward trend of unit sales starting in 2023. Exploring this insight further would include discovering the source of that growth and if the upward trend can be explained by other factors. In fact, data exploration is a continuous process. Insights generate new questions, which then generate even more insights. These observations are then interpreted and communicated to stakeholders during the last stage of the data analysis process.

[1]Few, Stephen. (2012). Show Me the Numbers: Designing Tables and Graphs to Enlighten. El Dorado Hills, CA: Analytics Press.

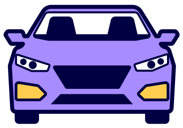
APPLYING CRITICAL THINKING 7.1: Think Critically During Data Exploration

Critical thinking is an integral part of data exploration:

- Exploring requires in-depth understanding of data relationships, how they are presented, and the insights they can generate. For example, many visualizations can present and explore time series, including area charts, column charts, line charts, sparkline charts, and waterfall charts. You must know which chart to use when. When exploring time series, you are looking for trends, cycles, and irregularities **(Knowledge)**.
- Data exploration is often pattern-based. You will look for patterns and structures you have encountered before and develop more that you can leverage in future analyses. One example is breaking trends down to better understand the reason for changes. If there is an upward trend in units sold, is this also true for all products and regions **(Self-Reflection)**?

Exploring Data with PivotTables

Data exploration investigates data from different angles to collect insights. A widely-used tool for this is the Excel PivotTable. You learned in the foundational data analysis skills chapter that a PivotTable can quickly rearrange data to help answer important business questions. Here, we illustrate the key elements of data exploration using a PivotTable.



Data The data set in **Illustration 7.2** (A) summarizes unit sales for Honda Motors North America (HNA) for the period 2021–2025 in a cross tabulation table.[2] Illustration 7.2 (B) shows the same data as a flat table.[3] Recall that for data analytics, flat tables are preferred over cross-tabulation tables because the column headers in flat tables do not contain data values useful for analysis purposes. The flat table in Illustration 7.2 (B) will be used to build PivotTables.

ILLUSTRATION 7.2 HNA Data Set

Country	Model	Type	2021	2022	2023	2024	2025
USA	Civic	Sedan	320981	301882	292331	291002	255423
USA	Accord	Sedan	243192	245998	231441	309885	344771
USA	CR-V	SUV	175883	160886	190001	220877	252019
USA	Pilot	SUV	140444	142980	139441	142917	125090
USA	Odyssey	Minivan	188664	167123	150872	150881	139009
USA	Ridgeline	Truck	58322	55897	56889	55899	54891
Canada	Civic	Sedan	230887	242998	275667	381998	480871
Canada	Accord	Sedan	195232	200872	210665	253988	319755
Canada	CR-V	SUV	135423	129809	126592	114119	98077
Canada	Pilot	SUV	40998	35672	39811	43125	47329
Canada	Odyssey	Minivan	25229	7761	26981	22099	19822
Canada	Ridgeline	Truck	19008	31887	42001	55089	88081

(A) Cross Tabulation Structure of HNA Data Set

Country	Model	Type	Year	UnitsSold
Canada	Accord	Sedan	2021	195232
Canada	Accord	Sedan	2022	200872
Canada	Accord	Sedan	2023	210665
Canada	Accord	Sedan	2024	253988
Canada	Accord	Sedan	2025	319755
Canada	Civic	Sedan	2021	230887
Canada	Civic	Sedan	2022	242998
Canada	Civic	Sedan	2023	275667
Canada	Civic	Sedan	2024	381998
Canada	Civic	Sedan	2025	480871

(B) Flat Structure of HNA Data Set

Illustration 7.3 (A) is a PivotTable made with Excel using the HNA data set. It lists HNA's models in descending order of total units sold. Illustration 7.3 (B) shows how the PivotTable in panel A is created using the PivotTable Fields dialog box, which is Excel's tool for defining a PivotTable's content.

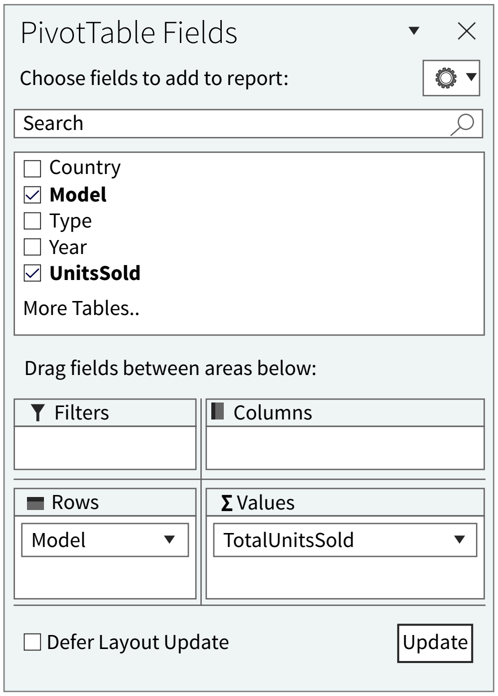
[2]The data set is fictitious.

[3]Illustration 7.2 (A) shows the full data set. Illustration 7.2 (B) shows a partial data set; the first 10 rows of the flat table.

ILLUSTRATION 7.3 Excel PivotTable and PivotTable Dialog Box

Models	TotalUnitsSold
Civic	3,074,040
Accord	2,555,799
CR-V	1,603,686
Odyssey	898,441
Pilot	897,807
Ridgeline	517,964
TotalUnitsSold	**9,547,737**

(A) PivotTable: Total Units Sold Per Model[4]



(B) PivotTable Creation

The five components used for data exploration with PivotTables are fields, values, rows, columns, and filters (Illustration 7.3 (B)).

Fields

The Fields area lists all the data elements available for exploration purposes. They can be dragged and dropped to other areas to build data relationships and filter the data. In Illustration 7.3, the Model and UnitsSold fields are used for exploration.

Values

The Values area in Illustration 7.3 (B) represents the number or numbers to be analyzed. It can be used to explore data in different ways:

- Drag and drop any field into the Values area and apply mathematical operations such as average, count, or sum to it.
- Create calculated fields.

Examples of accounting-related values that could be analyzed include gross revenue, net revenue, taxes, cost, profit, and more. For HNA, the values in the UnitsSold field are summed, generating the total number of units sold during the 2021–2025 period.

Rows and Columns

In Illustration 7.3 (B), the Model field has been dragged to the Rows area so the table will calculate and show the total units sold per model. The TotalUnitsSold column in Illustration 7.3 (A) can then be sorted and formatted.

When the Country field is added to the Columns area in Illustration 7.3 (B), the resulting crosstabulation table (**Illustration 7.4**) shows the total units sold per model, represented by the rows, and per country, which is represented by the columns.

[4]Some minor formatting was applied.

ILLUSTRATION 7.4 Excel PivotTable Total Units Sold per Model per Country

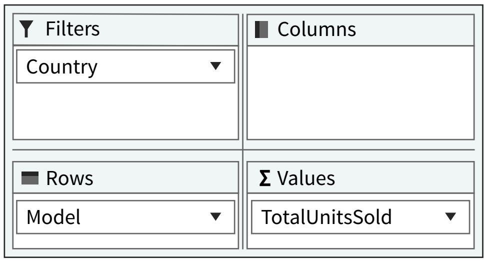
Total UnitsSold Models	Countries Canada	USA	TotalUnitsSold
Accord	1,180,512	1,375,287	2,555,799
Civic	1,612,421	1,461,619	3,074,040
CR-V	604,020	999,666	1,603,686
Odyssey	101,892	796,549	898,441
Pilot	206,935	690,872	897,807
Ridgeline	236,066	281,898	517,964
TotalUnitsSold	**3,941,846**	**5,605,891**	**9,547,737**

Any fields can be dragged to the Rows and Columns areas and dropped. For example, we can analyze the number of units sold per Type (Rows) per Country (Columns). It is also possible to drag and drop more than one field into the Rows and Columns sections to drill further down into the data.

Filters

Using filters further enhances data exploration with Excel PivotTables. Filters let us determine what data should be considered for analysis, and they can be created for any field. The PivotTable in **Illustration 7.5** (A) creates a filter for the Country field, while Illustration 7.5 (B) shows how to use this filter by selecting a value for the Country field. In this case, the selected value is USA.

ILLUSTRATION 7.5 Excel PivotTable Filters



(A) Creating a PivotTable Filter

Country	USA
Models	**TotalUnitsSold**
Accord	1,375,287
Civic	1,461,619
CR-V	999,666
Odyssey	796,549
Pilot	690,872
Ridgeline	281,898
TotalUnitsSold	**5,605,891**

(B) PivotTable with Filter Applied

Creating Relationships, Filtering, and Dragging and Dropping

The PivotTable definition in Illustration 7.3 (B) brings together some key techniques for data exploration–creating data relationships, filtering, and dragging and dropping.

First, data relationships can be created between the variables in the Values, Rows, and Columns areas and explored. Here are two examples of such data relationships:

- How can the Values variable number be broken down by the values of the Rows and Columns variables? For HNA, what is the share of each model (rows) in the total number of units sold (values)?
- How can the different models (rows) be ranked based on total units sold (values)? What are HNA's best-selling models?

There are some similarities between this discussion and the concept of information modeling, which was covered earlier in this course. The numbers in the Values area can be compared to measures, while the fields in the Rows and Columns areas can be compared to dimensions.

Next, filtering helps identify what data should be considered for analysis. An example of this is applying a filter that would exclude outliers in an analysis. Finally, the ability to drag and drop fields makes exploring different scenarios effortless. Here are some questions that can be answered by dragging and dropping fields using the HNA data set in a PivotTable:

- What is the total number of units sold (values) per country (rows) per model (columns) and how do the numbers compare?
- What is each country's share (rows) in the total number of units sold (values)?
- How do USA (filter) sales, in terms of total units sold (values), compare across the different models (rows)?

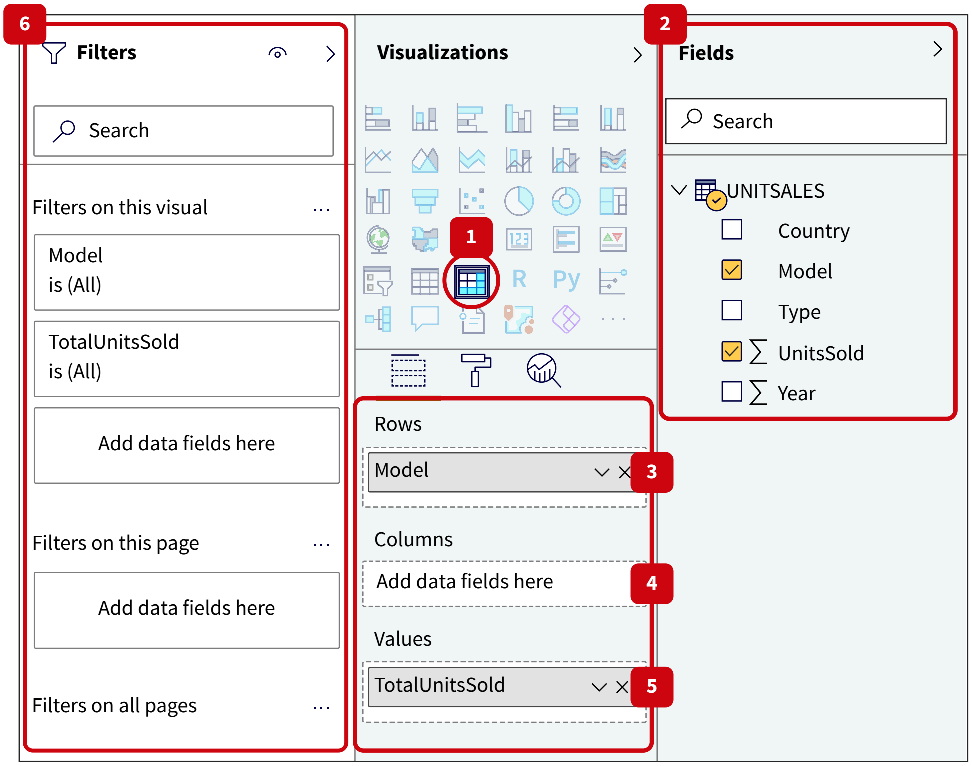
Data Exploration Across Tools

Excel is not the only business intelligence software that uses data relationships, filtering, and drag and drop for data exploration. **Illustration 7.6** (A) is a Power BI pivot table that summarizes HNA's total units sold per model during the last five years. In Power BI a pivot table is called a matrix. The second part of the illustration (B) shows how it was created.[5]

ILLUSTRATION 7.6 Creating a Power BI Matrix

Model	TotalUnitsSold
Accord	2,555,799
Civic	3,074,040
CR-V	1,603,686
Odyssey	898,441
Pilot	897,807
Ridgeline	517,964
Total	**9,547,737**

(A) Power BI Matrix



(B) Power BI Matrix Creation

Notice the similarity between creating an Excel PivotTable in Illustration 7.3 (B) and a Power BI matrix in Illustration 7.6 (B):

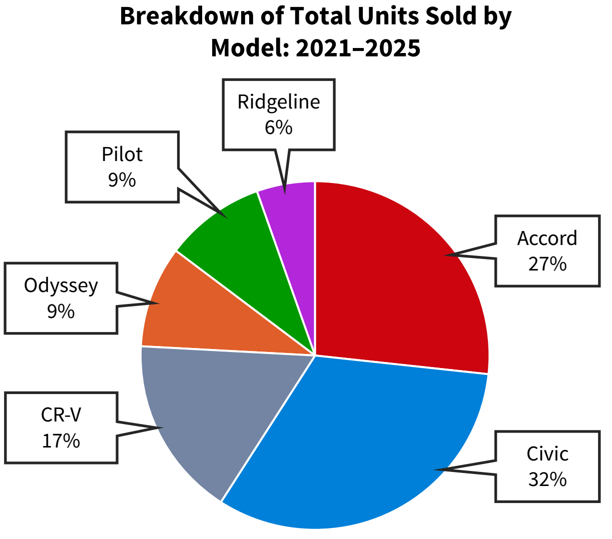
1. In Power BI, we can choose between many drag and drop visualizations. In this example, matrix was selected.
2. The Fields pane corresponds to the fields list in the PivotTable Fields dialog box.
3. The Rows slot in the Fields tab of the matrix visualization corresponds with the Rows area in an Excel PivotTable definition.
4. The Columns slot in the Fields tab of the matrix visualization corresponds with the Columns area in an Excel PivotTable definition.

[5]We used Power BI's matrix visualization. However, since no columns are defined, we could have used the table visualization.

5. The Values slot in the Fields tab of the matrix visualization corresponds with the Values area in an Excel PivotTable definition.
6. The Filters pane is an enhanced version of the Filters area in an Excel PivotTable definition.

Charts are often more effective than tables for analysis. Power BI charts, which are visualizations, represent one or more data relationships and have a drop and drag structure. In Excel it is easy to convert PivotTables into charts with the PivotChart tool. PivotCharts graphically represent the data relationships in PivotTables. **Illustration 7.7** is a pie chart for the PivotTable in Illustration 7.3 (A). It highlights each model's share in the total units sold, a part-to-whole relationship, which is discussed later.

ILLUSTRATION 7.7 Excel PivotChart



Next, let's examine how charts can represent and explore data relationships.

Apply It 7.1
Explore Data with PivotTables

Data **Managerial Accounting** Happy Colors is a small paint company that prepares paint cans that are sold to retailers. CEO Mika Jacobson supervises three employees, and there are three types of products–low, high, and premium paint. Regardless of the type of paint, it should take about three minutes to prepare a can. The cost of a can depends on the type of product:

- Low: $5
- High: $10
- Premium: $14

Since defective paint cans cannot be sold, Mika inspects each can herself. Losses resulting from defective cans are an important issue for Happy Colors, so Mika hires your firm to help understand this issue. Here is the data dictionary for this data set.

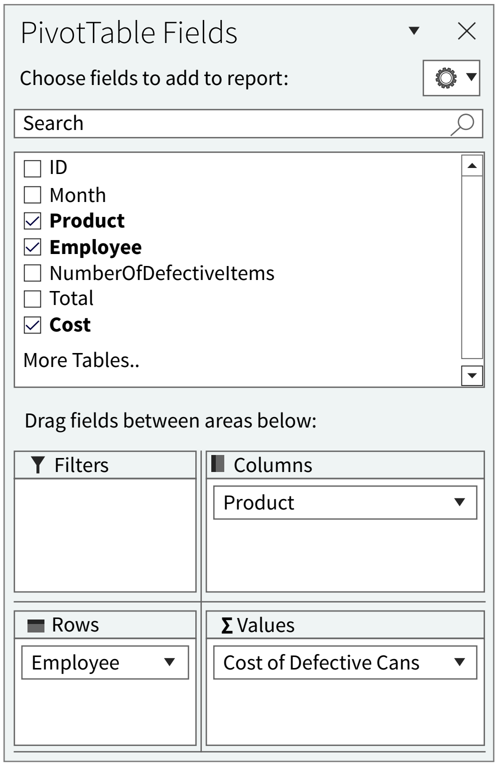
Data Dictionary for Happy Colors

Name	Description
ID	A unique ID for each recording of an employee's output per month per product type.
Month	The name of the month.
Product	The name of the product type: low, high, or premium.
Employee	The name of the employee.
NumberOfDefectiveItems	The number of defective cans by an employee for a product type in a specific month.
Total	The total number of cans by an employee for a product type in a specific month.
Cost	The total loss (in dollars) resulting from the detective items by an employee for a product in a specific month.

1. Create a PivotTable that helps you understand the losses per employee per product type.
2. What insights does the PivotTable generate?

SOLUTION

1. Using Excel, the PivotTable can be created by selecting the product, employee, and cost fields. Drag the Cost field to Values, the Employee field to Rows, and the Product field to Columns.



After applying some formatting, the resulting PivotTable looks like this:

Cost of Defective Cans	**Product Type**			
Employee	**High**	**Low**	**Premium**	**Total**
Andre	$ 30,500	$ 20,750	$ 1,530	$ 52,780
Bianca	$ 24,750	$ 24,250	$ 1,147	$ 50,147
Camille	$ 35,500	$ 14,750	$ 825	$ 51,075
Total	**$ 90,750**	**$ 59,750**	**$ 3,502**	**$ 154,002**

2. The table shows that the losses resulting from defective items are somewhat similar for the three employees. However, Andre has the highest losses for premium paint cans, Bianca has the highest losses for low paint cans, and Camille has the highest losses for high paint cans.

7.2 How are Data Relationships Visualized for Exploration?

LEARNING OBJECTIVE 2

Explore foundational data relationships through visualizations.

You have learned that a data relationship describes how data elements, or values, relate to each other. Each data analytics project will have a unique set of data relationships. While there is no single, best approach for investigating data relationships, a structured set of data exploration patterns can help answer questions about which data relationships can be explored, how they can be illustrated in visualizations, how to create those visualizations, and what they can tell us.

There are two types of data exploration patterns. Some patterns explore a foundational data relationship with a single visualization, while others explore data by integrating data relationships. This section presents eight foundational data exploration patterns. Each pattern identifies and describes the data relationship and visualizations that could represent it. The pattern also shows how to explore the relationship and the insights that can be collected from doing so.

APPLYING CRITICAL THINKING 7.2: Use Patterns to Explore Data

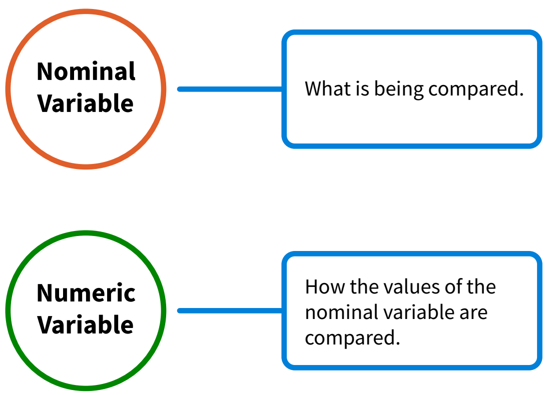
Data exploration patterns are *real-world* best practices for using data relationships for exploration. They describe which data relationships exist and how to effectively visualize and explore them **(Knowledge)**:

- Data relationship: Time series data relationships compare observations over time.
- Visualization: Line charts are considered the most effective visualization for exploring time series. Best visualization practices include starting the line chart's y-axis at zero.
- Exploration: Line charts generate insights by identifying trends, cycles, and irregularities.

Data Exploration Pattern 1: Nominal Comparison

A **nominal comparison data relationship** compares the values of a nominal variable based on the values of a second, numeric variable. **Illustration 7.8** shows its **exploration structure**, which is a visual that describes the different elements used in the exploration and how they are related. For nominal comparisons, the exploration structure is a nominal variable, which is what is being compared, and a numeric variable, which is how the values of the nominal variable are compared.

ILLUSTRATION 7.8 Exploration Structure for a Nominal Comparison Data Relationship



Visualizations

Visualizations for nominal comparisons include bar charts, column charts, dot plots, and lollipop charts. Let's go back to the Honda North America example from earlier in the chapter. **Illustration 7.9** is a column chart that compares HNA's models, which is the nominal variable, using the number of units sold, which is the numeric variable. Each model is presented by a column (x-axis), and the length of the column is determined by the number of units sold for that model (y-axis).

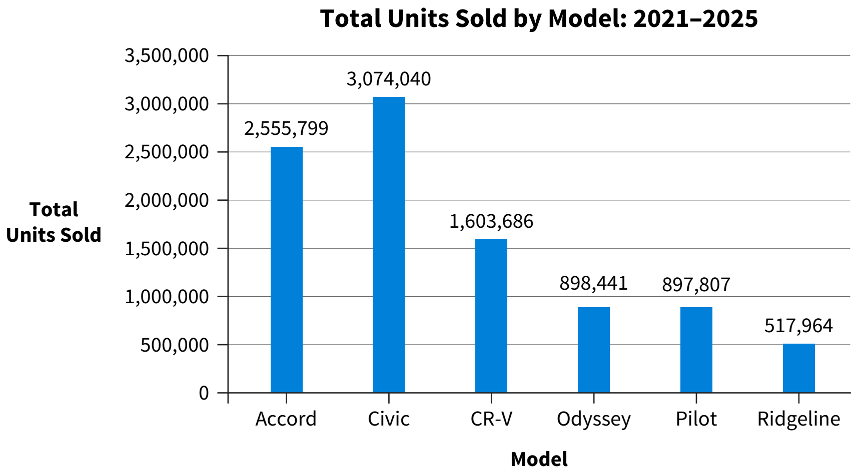


ILLUSTRATION 7.9 Visualizing a Nominal Comparison Data Relationship with a Column Chart

The Excel PivotTable used to create this column chart is shown in **Illustration 7.10**.

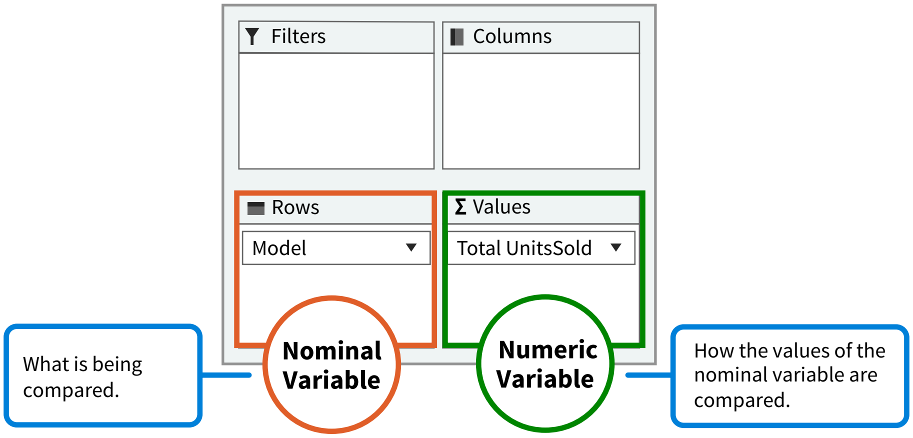


ILLUSTRATION 7.10 Excel PivotTable Definition

Exploration and Insights

A nominal comparison can quickly evaluate data and collect initial insights, which is especially useful when working with a new data set. It allows us to compare the sizes of each category–which is the biggest, which is the smallest, if one category is twice as big as another, and more. Illustration 7.9 shows which HNA models sold the most and least units in the 2021–2025 period. The same chart also indicates if unit sales across models are similar or if there are significant differences.

You have learned that dragging and dropping different variables in different areas is key to data exploration. Dragging and dropping variables in Illustration 7.10 makes the opportunities for nominal comparison endless:

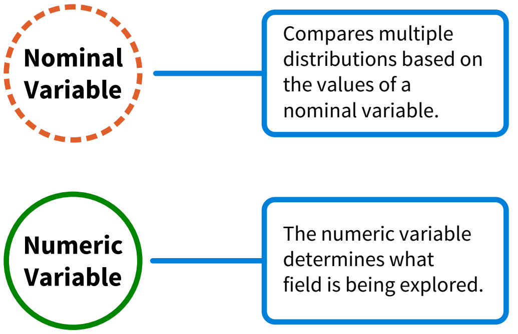
- Any nominal variable, or *what* is being analyzed, can be compared using any numeric variable, or *how* the nominal variable is being analyzed.
- Filters can also be applied to both the nominal variable and the numeric variable.

What if an accounting firm regularly analyzes the performance of its staff members? How would they do this? First, the staff members (*what*) are being compared, and they would be selected with a filter. We could use many measures (*how*) to compare them, including total hours worked, total revenue generated, number of clients, average revenue per client, and average client response time. All these variables can be dragged to the numeric variable slot labeled Values. This is why the information modeling step is important for data exploration–it allows creating additional measures that you can later use in analysis.

Data Exploration Pattern 2: Distribution

A **distribution data relationship** (**Illustration 7.11**) shows how the values of a numeric variable are distributed, or spread out, by providing the lowest value, the highest value, the median, the interquartile range, and so on.

ILLUSTRATION 7.11 Exploration Structure for a Distribution Data Relationship



A common scenario, shown in Illustration 7.11, is to create and compare multiple distributions of the same numeric variable based on the different values of a nominal variable. The dotted line indicates that such a variable is valuable for exploration, but optional when defining a distribution data relationship.

Visualizations

Several visualizations portray distributions, including histograms, violin plots, and box-and-whisker charts (or boxplot charts). Box-and-whisker charts are both powerful and detailed, so they are used here to illustrate a visualization for this pattern.

Let's go back to the HNA example. **Data** The company has a spreadsheet from one of its New York dealerships. They want to know how the profits generated by selling cars vary. **Illustration 7.12** summarizes the spreadsheet data about the cars sold by the dealership last December.

ILLUSTRATION 7.12 Data Structure of the Dealership Profit Data

Field Name	Description
ID	The internal ID that identifies the car.
Model	The car's model.
Profit	The profit made when selling the car is selling price minus dealer invoice. This is also known as front-end gross profit.

Illustration 7.13 shows the first ten rows of the data set.

ID	Model	Profit
1	Civic	$ 2,125
2	CR-V	$ 2,798
3	Odyssey	$ 2,144
4	Accord	$ 2,749
5	Accord	$ 1,810
6	Accord	$ 1,844
7	Odyssey	$ 4,446
8	Civic	$ 1,912
9	Pilot	$ 3,107
10	Civic	$ 2,101

ILLUSTRATION 7.13 First Ten Rows of the Dealership Profits Data Set.

Illustration 7.14 shows a box-and-whisker chart for the Profit field. The red circle represents the average. The bottom of the green box is the first quartile, and the top of the orange box is the third quartile. The median is the intersection between the green and orange boxes, and the length of the green and orange boxes combined is the interquartile range. Finally, the lower fence portrays the minimum value, while the upper fence portrays the maximum value.

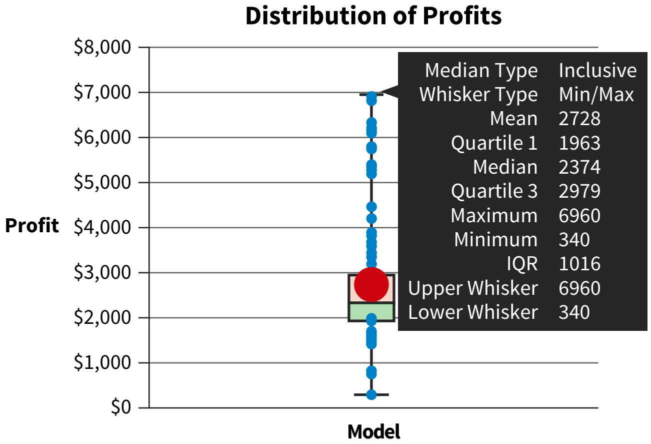
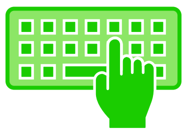


ILLUSTRATION 7.14 Visualizing a Distribution Data Relationship with a Box-And-Whisker Chart

Illustration 7.15 shows how this chart would be created in Power BI. Power BI requires dragging a field into the Axis slot. The values for this field represent the data points. Dragging a primary key into this slot is good practice because all data points and their unique values are then represented. A similar box-and-whisker chart can also be created in Excel. (**Data** See How To 7.1. at the end of the chapter to learn how to do this.)

How To



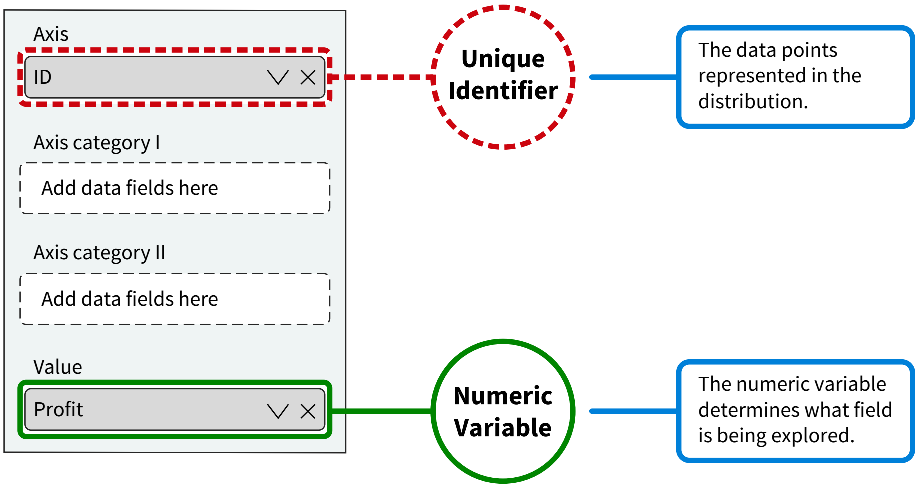


ILLUSTRATION 7.15 Creating a Box-and-Whisker Chart in Power BI

As mentioned earlier, creating and comparing multiple distributions of the same numeric variable based on the different values of a nominal variable is common in this kind of analysis. **Illustration 7.16** shows how to make a chart with this comparison using Power BI:

- Drag Model to the Axis category I slot.
- A separate distribution is then created for each model.

ILLUSTRATION 7.16 Creating a Box-and-Whisker Chart with Multiple Distributions

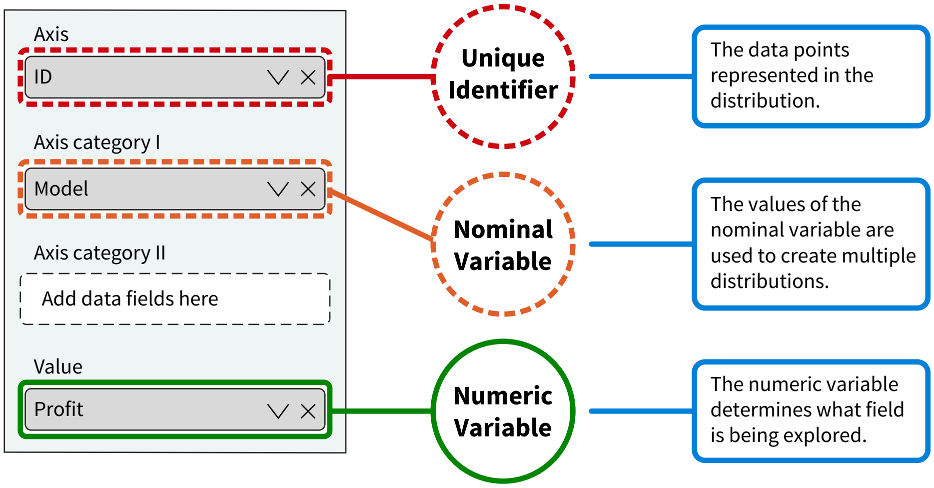
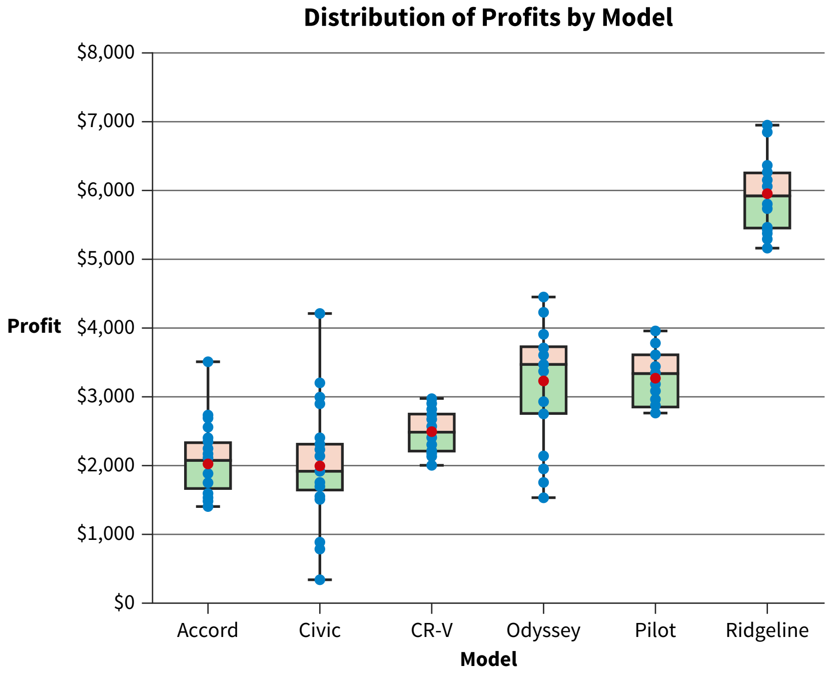


Illustration 7.17 is the resulting chart.

ILLUSTRATION 7.17 Parallel Profit Distributions for HNA Car Models



Exploration and Insights

Dragging and dropping variables also supports data exploration using this pattern. Using the exploration structure for a distribution data relationship, we can quickly examine the distribution of any numeric field in a data set and use any nominal field in the data set to compare data distributions.

Illustration 7.14 showed the distribution information provided by Power BI, including the median, the mean, the interquartile range (IQR), the lowest value, and the highest value. This makes it possible to investigate outliers, skewness, and more. The chart in Illustration 7.17 allows even more comparison across distributions:

- The profits for the Ridgeline model are higher than for the other models.
- The profits for the CR-V and Pilot models are consistent, which means there is little variation.
- The profits for the Odyssey and especially the Civic models are more spread out.

It might be necessary to explore the data more to better understand the source of the variation. The source could be a result of the higher price for a specific model, the number of options available for a specific model, the negotiation skills of the salespeople, or sales promotions, for example.

Distribution analysis has many applications, including analyzing employee compensation. To understand how salaries are distributed, distribution analysis could be used to answer the following questions:

- Are salaries right-skewed? Are the higher salaries more spread out?
- How do salaries compare across departments?
- How does an organization's compensation distribution compare to the distribution for other organizations?

Data Exploration Pattern 3: Deviation

A **deviation data relationship** shows how a set of actual values deviate from their reference values, which are budgeted or forecasted values. Deviation relationships are everywhere in accounting. Variance analysis is a good example, such as the difference between actual and standard costs. The exploration structure for deviation relationships shown in **Illustration 7.18** contains the variable being compared and the variables being used for comparison purposes–actual, target, and deviation.

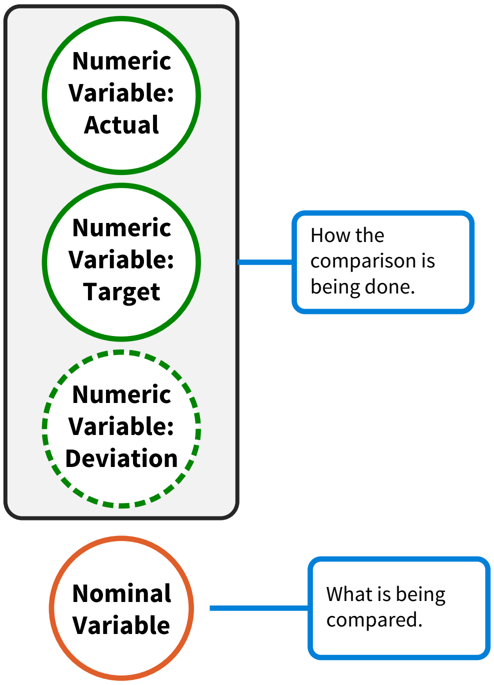


ILLUSTRATION 7.18 Exploration Structure for a Deviation Data Relationship

Visualizations

Clustered bar and column charts (used in the following example), gauges, bullet charts, and more can be used for deviation analysis. Imagine that HNA would like to explore whether each vehicle model met its sales expectations. **Data** The company has another data set that includes both the actual and the budgeted unit sales for the 2025 period (**Illustration 7.19**).

ILLUSTRATION 7.19 HNA Budgeted Unit Sales-2025

Country	Model	Type	ActualUnits	BudgetedUnits
US	Civic	Sedan	255423	300000
US	Accord	Sedan	344771	350000
US	CR-V	SUV	252019	25000
USA	Pilot	SUV	125090	160000
USA	Odyssey	Minivan	139009	145000
USA	Ridgeline	Truck	54891	55000
Canada	Civic	Sedan	480871	425000
Canada	Accord	Sedan	319755	275000
Canada	CR-V	SUV	98077	110000
Canada	Pilot	SUV	47329	45000
Canada	Odyssey	Minivan	19822	30000
Canada	Ridgeline	Truck	88081	65000

This data set was used to create the clustered column chart in **Illustration 7.20** with Power BI.

ILLUSTRATION 7.20 Visualizing a Deviation Data Relationship with a Clustered Column Chart

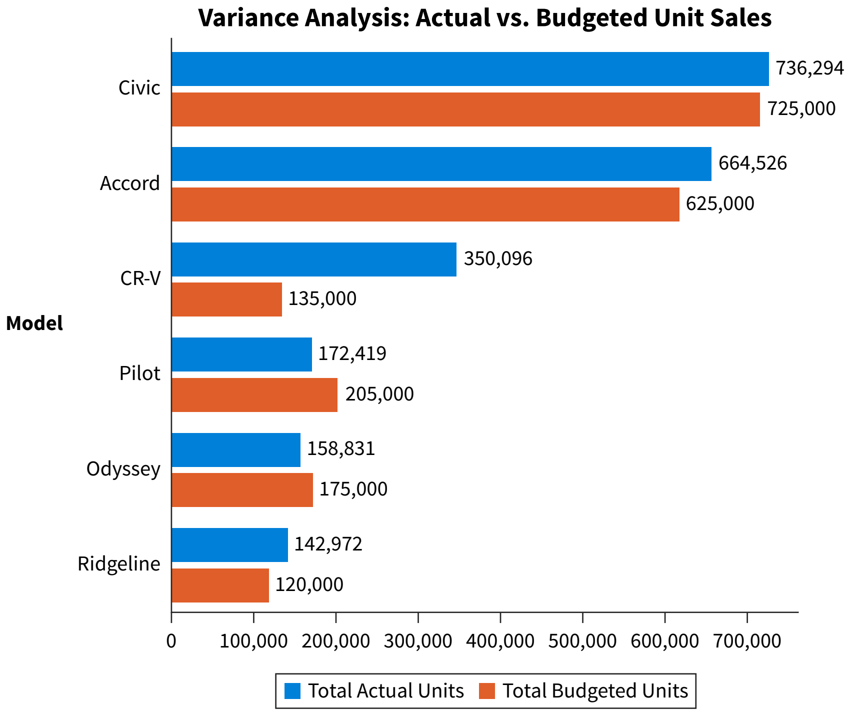
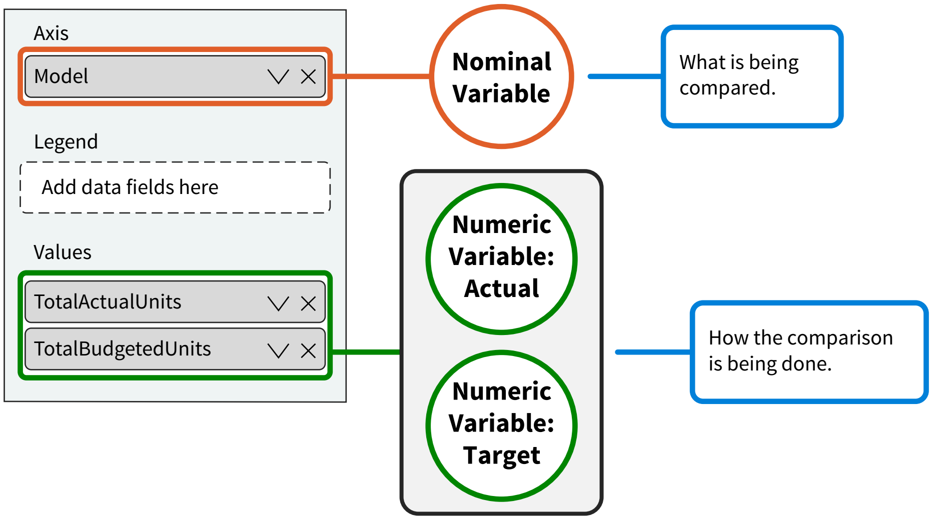


Illustration 7.21 shows how the chart was created.

ILLUSTRATION 7.21 Creating a Clustered Column Chart in Power BI



Unlike a column chart, a clustered column chart uses multiple variables for comparison purposes. While the clustered column chart can make comparisons across models, it also provides a comparison within a model. For example, the comparison in Illustration 7.20 is between the actual number of units sold and the budgeted number of units sold, which are the two variables in the Values area:

- TotalActualUnits represents the sum of all values in the ActualUnits field.
- TotalBudgetedUnits represents the sum of all values in the BudgetedUnits field.

Exploration and Insights

Additional variables can be dragged into the Values slot in Illustration 7.21. For example, variance, a variable that is calculated as the difference between the actual and the budgeted units, could be added. There are also three nominal variables in this data set to choose from: country, type, or model.

Different types of insights can be collected from deviation analysis, including when results are higher or more favorable than expected, lower or less favorable than expected, and if targets are unreasonable. Once we identify these insights, the next step might be exploring possible causes, such as how to reverse unfavorable deviations. This is particularly useful when the variances are significant.

Variances have historically been valuable tools for accountants to indicate issues that should be addressed:

- Inefficiencies such as inexperience in negotiating prices, ill-qualified employees, employees assigned the wrong task, use of low-quality resources, or inadequate training.
- Unexpected changes such as changes in prices, or a decrease in demand.
- Poor budgeting.

A possible insight that can be collected from Illustration 7.20 is that most models have a favorable variance, but some (Pilot and Odyssey) have an unfavorable variance. Another is that, except for the CR-V model, most variances are insignificant.

Data Exploration Pattern 4: Ranking

A **ranking data relationship** (**Illustration 7.22**) orders the values of a variable sequentially based on the values of a second variable. Ranks are determined by some quality, such as highest, lowest, fastest, slowest, as defined by the second variable. The ranking can be displayed in ascending or descending order. It is also possible to calculate a rank explicitly.

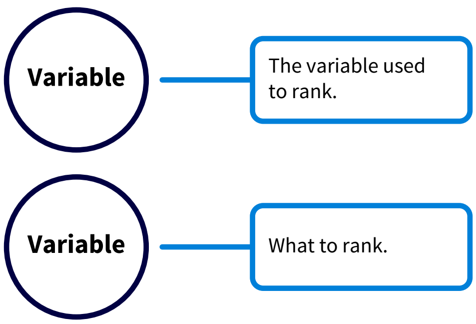
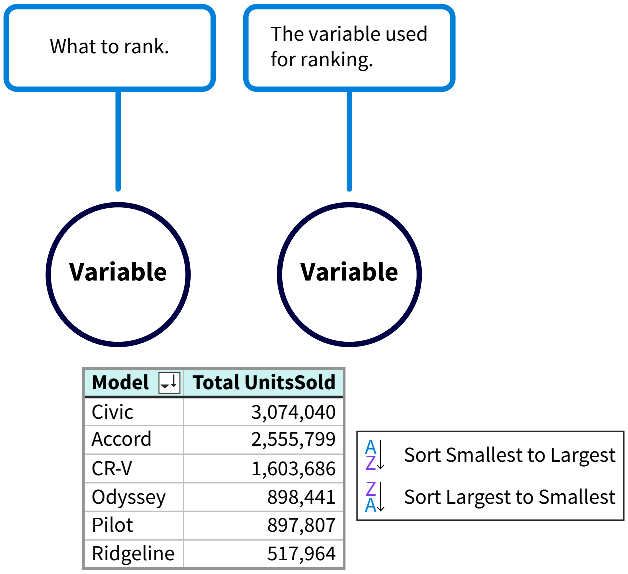


ILLUSTRATION 7.22 Exploration Structure for a Ranking Data Relationship

Visualizations

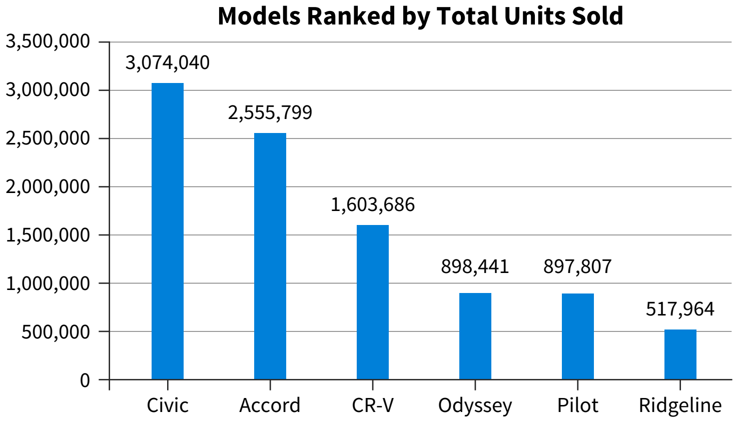
There are no specific ranking visualizations, but many integrate ranking information, including tables, bar charts, and column charts. **Illustration 7.23** (B) shows a column chart that ranks HNA's models (*what to rank*) based on the units sold in the 2021–2025 period (*how to rank*). Illustration 7.23 (A) shows this ranking relationship created for a PivotTable in Excel.

ILLUSTRATION 7.23 Integrating a Ranking Data Relationship into a Column Chart



Model	Total UnitsSold
Civic	3,074,040
Accord	2,555,799
CR-V	1,603,686
Odyssey	898,441
Pilot	897,807
Ridgeline	517,964

(A) Creating a Ranking Relationship for a PivotTable



(B) Integrating a Ranking Relationship with a Column Chart

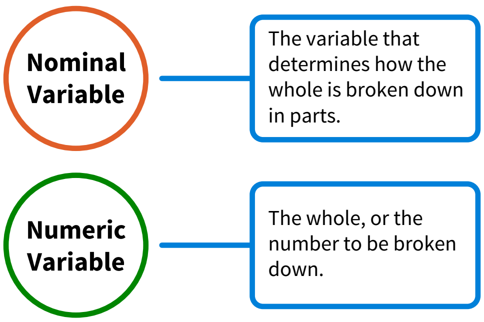
Exploration and Insights

Different ranking relationships can be explored by dragging and dropping. Customers can be ranked based on their sales amount or their age; employees can be ranked based on their salaries or their seniority. Measures, such as total outstanding debt of at least 60 days, can also be used to rank customers. Ranking can determine, for example, the top 25 or bottom 25 customers based on revenue. The column chart in Illustration 7.23 indicates a model's rank based on units sold during the 2021–2025 period. The Civic is the best-selling model, followed by the Accord.

Data Exploration Pattern 5: Part-to-Whole

A **part–to-whole data relationship** (**Illustration 7.24**) compares parts to wholes and looks at how the different parts compare to each other. For example, an organization might generate $100,000 in sales, which is the whole. The home, office, and garden product categories generate $30,000 (30 %), $50,000 (50%), and $20,000 (20%), respectively. These three product categories represent the parts.

ILLUSTRATION 7.24 Exploration Structure for a Part-To-Whole Data Relationship



A part-to-whole visualization should define the number that is the whole and how the whole should be broken down into parts.

Visualizations

Multiple visualizations can model part-to-whole relationships, including pie charts, donut charts, stacked bar charts, stacked column charts, and treemaps. While treemaps are considered superior for portraying part-to-whole relationships, here the more commonly used pie chart is shown. The pie chart in **Illustration 7.25** shows the relative importance of each of HNA's models, which are the parts, in terms of the whole, which is the units sold during the 2021–2025 period. The relative importance is expressed as a percentage, or share.

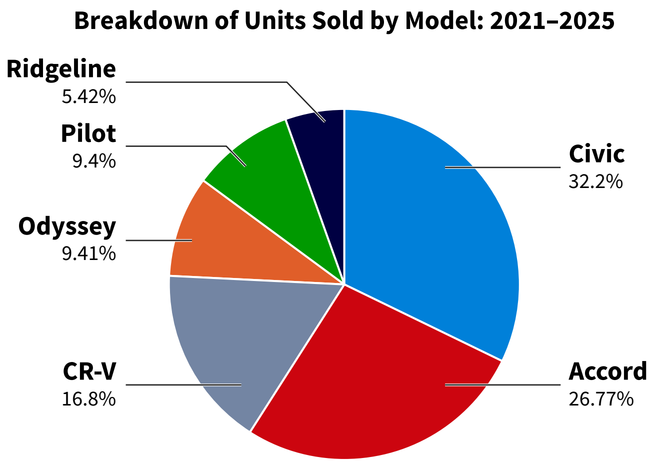


ILLUSTRATION 7.25 Visualizing a Part-To-Whole Data Relationship With a Pie Chart

Illustration 7.26 shows how this pie chart is created using Power BI:

- The Values slot defines the *what*: This refers to the number that is broken down. Here, it is the TotalUnitsSold measure.
- The Legend slot defines the *how*: This refers to how to break down the whole. Here, this is done by model. For each value of model the number of units sold is calculated, determining the size of its share of the pie. This visual shows the size of its share as a percentage.

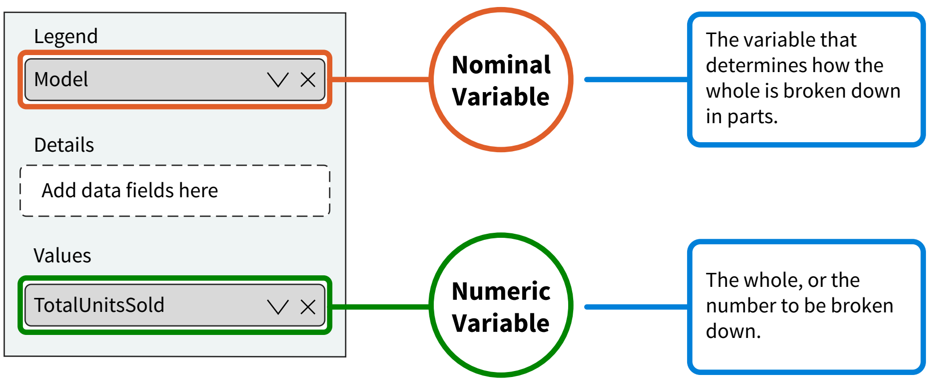


ILLUSTRATION 7.26 Creating a Pie Chart in Power BI

Exploration and Insights

Any measure can be dragged to the Values (*whole*) slot in Illustration 7.26. Measures tailored to specific needs can also be created, such as July sales to non-U.S. customers. Any nominal variable can be dragged to the Legend slot (*parts*).

Part-to-whole relationships are often used in accounting. Examples include the breakdown of expenses in multiple categories such as administrative expenses, depreciation, marketing expenses, and the breakdown of costs in terms of materials, labor, and overhead.

The pie chart in Illustration 7.25 shows how each model contributes, in relative terms, to the total units sold in the 2021–2025 period. It further shows that the Accord and Civic models combined count for more than 50% of the unit sales. The share of other models, such as the Ridgeline, is relatively small.

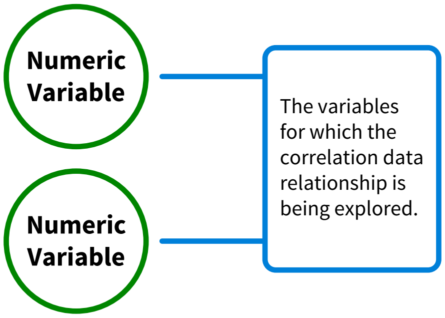
Data Exploration Pattern 6: Correlation

A **correlation data relationship** indicates the degree to which two variables move in the same or the opposite direction. For example, if the marketing expenses for a product increase, then it is likely that the sales for the same product also increase. There are two key features to consider with these relationships. The first is direction:

- The direction is positive if both variables move in the same direction.
- The direction is negative if both variables move in opposite directions.

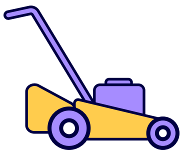
The second feature is the strength of the correlation. Strength indicates the degree of correlation between the two variables, ranging from no correlation to perfect correlation. **Illustration 7.27** shows the exploration structure of a correlation data relationship between numeric variables.

ILLUSTRATION 7.27 Exploration Structure for a Correlation Data Relationship



Visualizations

The most common visualization for exploring correlation is the scatterplot, which is also referred to as a scatter chart. It plots the coordinates for two variables for each data point.



Let's illustrate a scatterplot with an example from a new company. Buzz Cut is a landscaping business with two owners. These small business owners continuously face tough strategic, financial, and operational decisions, such as price estimates for properties. Data They have a spreadsheet that contains data about the size and average cutting time, in minutes, for all their properties (**Illustration 7.28**).

Price estimation is a critical decision for all landscape businesses. Buzz Cut uses the following decision model to make price estimates:

- It takes one minute to mow 1,000 square feet, and Buzz Cut charges a dollar per minute ($60 per hour). They would charge $50 for mowing a 50,000 square foot property.
- The minimum price per cut is $25.
- Reduced rates are offered for large lots.

All prices are rounded up: $25, $30, $35, and so on. In some cases, Buzz Cut's owners negotiate with the customers. Price estimation decisions rely on two assumptions:

1. The size of a lawn determines the time needed to cut it.
2. Time determines the cost and thus price.

ILLUSTRATION 7.28 First Ten Rows of the Property Data Size Time Spreadsheet

Property	Size	Time
1	4500	9
2	3000	6
3	50000	50
4	25000	45
5	4000	8
6	18000	18
7	21000	17
8	100000	120
9	22000	55
10	23000	25

The first assumption can be rephrased: "There is a strong positive correlation between lawn size and time: the bigger the lawn, the more time it will take to mow it."

Correlation is a data relationship. Strong correlation and positive direction are assumptions regarding the data relationship made by Buzz Cut. The scatterplot in **Illustration 7.29** explores the validity of these assumptions.

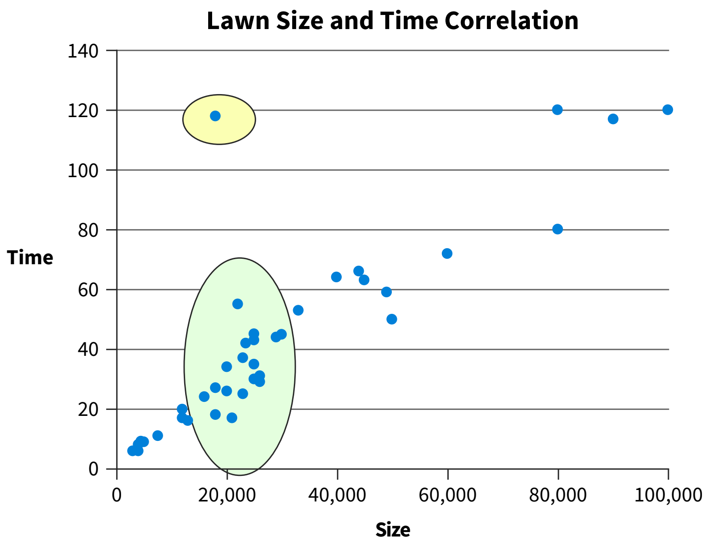


ILLUSTRATION 7.29 Scatterplot Showing Correlation Between Size and Time

This scatterplot was created in Power BI (**Illustration 7.30**).

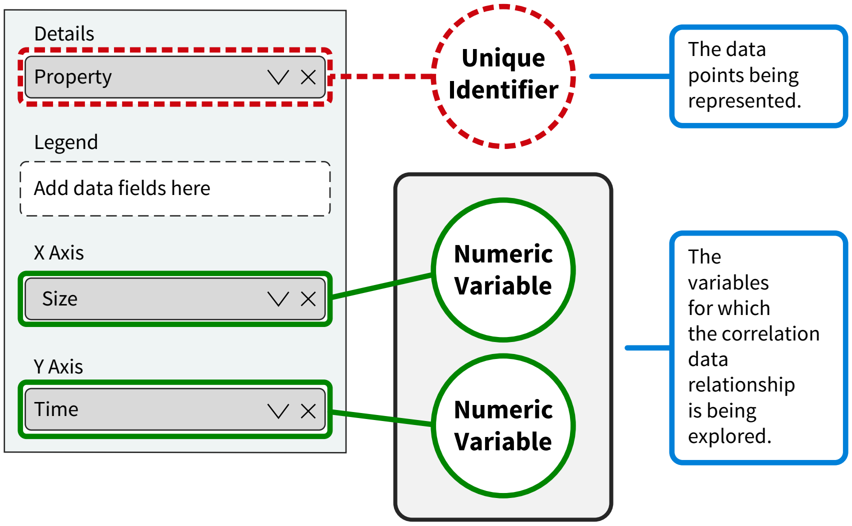


ILLUSTRATION 7.30 Creating a Scatterplot in Power BI

Exploration and Insights

Scatterplots are created by dragging and dropping numeric variables from a data set into the two areas shown in Illustration 7.30. Several insights could be collected once the scatterplot is created. For instance, there is a strong, positive correlation between lawn size and time. There are also two additional insights:

1. The data point with a yellow background shows an outlier. Talking to the owners of Buzz Cut reveals this is the result of a data entry issue. It takes 18 minutes to cut the property, but "118 minutes" was recorded.
2. The chart indicates that there might be other factors than just size influencing time. The data points with a green background in Illustration 7.29 indicate properties that are approximately the same size but require different amounts of time, with times ranging between 17 and 55 minutes (ignore the outlier here).

While the data do not indicate what the other factor or factors might be, the owners mention that properties with many trees, plants, and corners take more time to mow. This means that difficulty, or complexity, is another factor that influences time and cost, and the decision model should be adjusted to reflect this.

APPLYING CRITICAL THINKING 7.3: Evaluate Your Assumptions

The Buzz Cut scatterplot example shows that critical thinking is an integral part of data exploration:

- Unrealistic assumptions might result in insights that pose a significant risk. Underpricing might result in losses **(Risks)**.
- These insights could negatively affect all stakeholders. If there are other factors that drive time, employees could be evaluated unfairly, and underpricing might mean that Buzz Cut cannot pay creditors **(Stakeholders)**.

Data Exploration Pattern 7: Time Series

A **time series data relationship** defines the values of a variable at sequential points in time (**Illustration 7.31**).

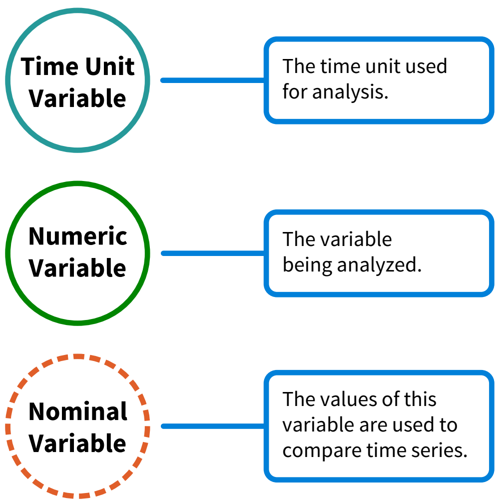


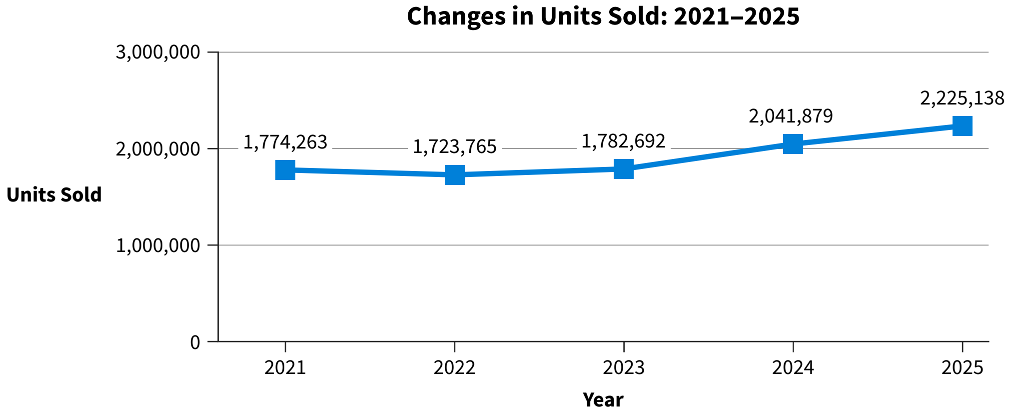
ILLUSTRATION 7.31 Exploration Structure for a Time Series Data Relationship

To display a time series, a visualization requires a time unit such as a minute, hour, day, or week, and a variable that changes over time.

Visualizations

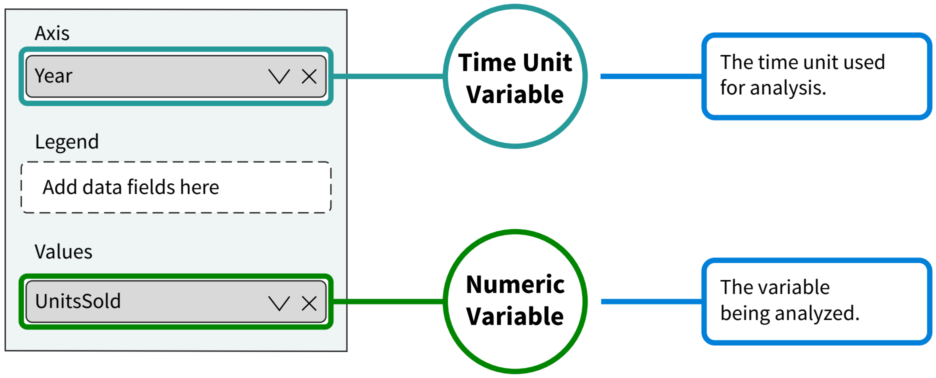
A line chart is the most common visualization for time series, but bar charts, column charts, area charts, waterfall charts, and sparkline charts can also be used. Let's go back to the Honda Motors North America (HNA) example. **Illustration 7.32** is a Power BI line chart that explores how HNA's unit sales changed during the 2021–2025 period.

ILLUSTRATION 7.32 Line Chart Showing Changes in HNA's Unit Sales During the 2021–2025 Period



It was created in Power BI by using year as the time unit and total unit sales as the numeric variable (**Illustration 7.33**).

ILLUSTRATION 7.33 Creating a Line Chart in Power BI



Exploration and Insights

Look for trends, cycles, and irregularities when exploring time series:

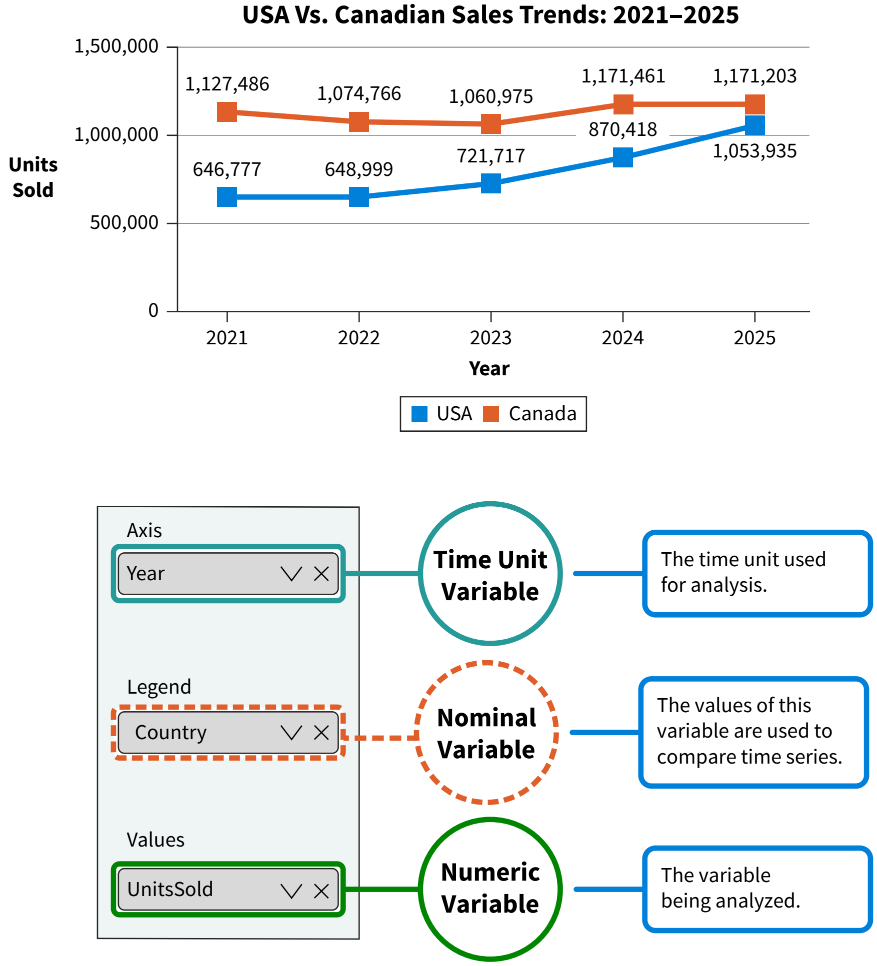
- A trend indicates the general direction (upward or downward) in which a variable moves over time.
- A cycle indicates a pattern of changes, which can vary in length and intensity over time. Seasonality refers to intra-year cycles with a fixed nature.
- Irregularities are unsystematic, typically short-term, fluctuations.

The time series in Illustration 7.32 shows a strong upward trend for HNA's unit sales starting from 2023. Dragging and dropping can explore this trend in more detail by comparing the trends for Canada and the USA. This new line chart (A) and how it was made (B) are shown in **Illustration 7.34**. This chart shows a very different picture: U.S. sales are flat, and growth is a result of increasing sales in Canada. Additional insights might be generated by also comparing trends for units sold by model or type.

ILLUSTRATION 7.34 Trend Analysis Units Sold by Country

(A) Trend Analysis: Units Sold by Country

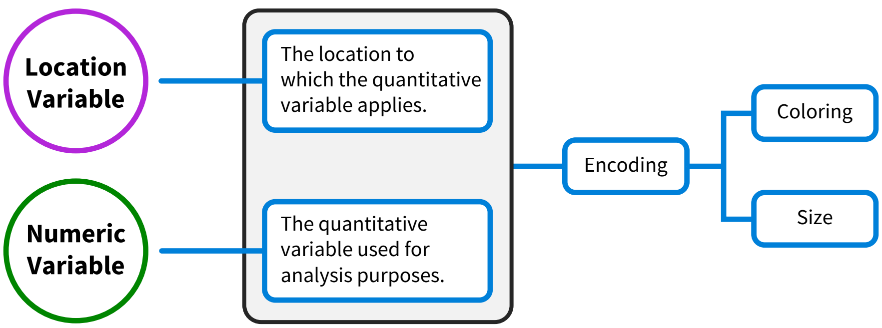
(B) Creating a Line Chart with Multiple Time Series



Data Exploration Pattern 8: Geospatial

In a **geospatial data relationship**, numeric values are assigned to locations and encoded through coloring or shading and the size of the bubbles within the visualization (**Illustration 7.35**).

ILLUSTRATION 7.35 Exploration Structure for Geospatial Data Relationships

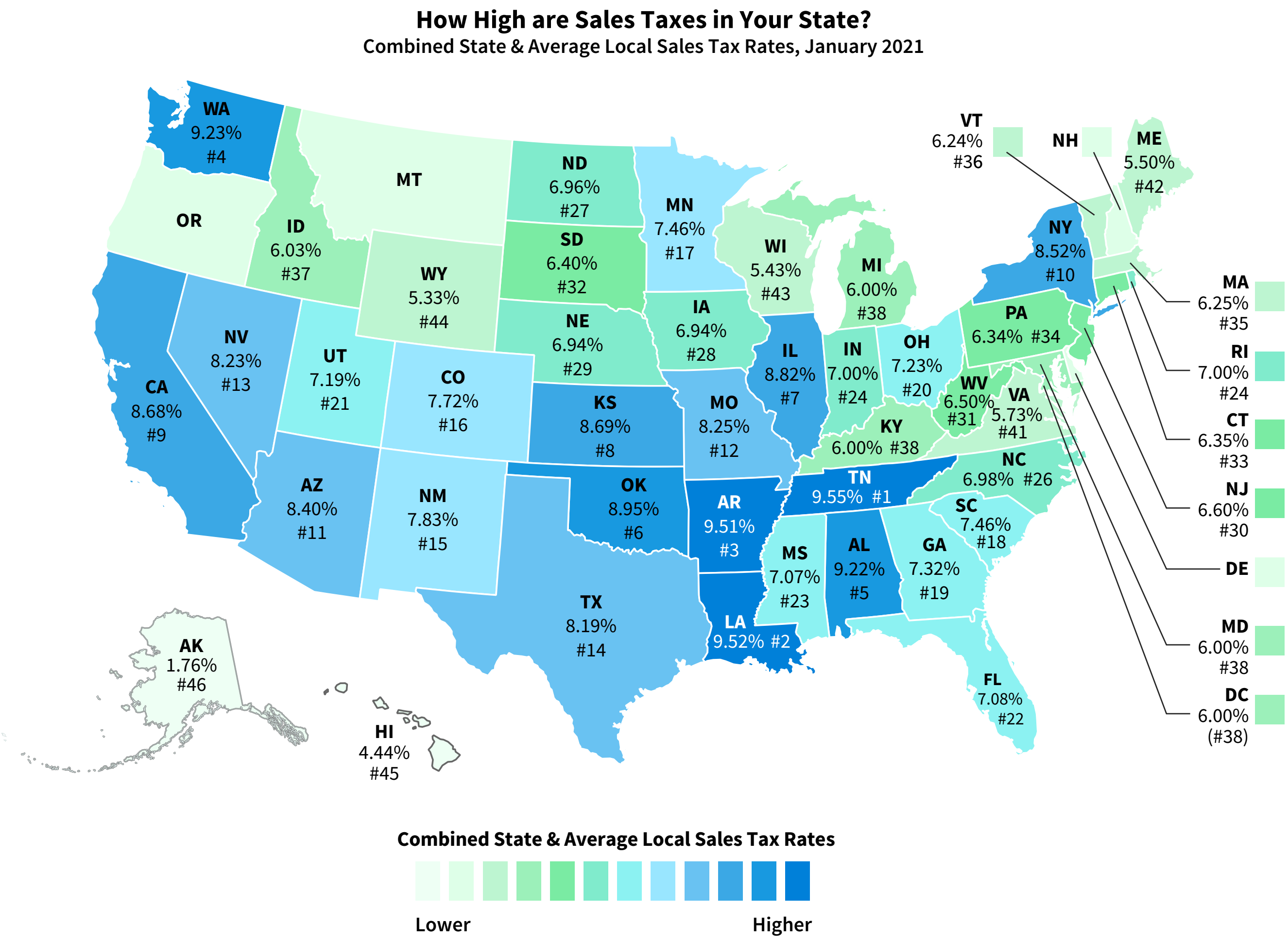


Visualizations

Geospatial relationships are defined using maps:

- A choropleth map uses color intensity to represent data values. In **Illustration 7.36**, color intensity differentiates among the combined state and average local sales tax rates for U.S. states. In this map, state is the location variable, tax rate is the numeric variable, and encoding is based on color intensity.
- A proportional symbol map uses symbols—often bubbles/circles—and the size of the symbol represents the data value. The larger the symbol, the higher the value. A business could create a map that shows the total revenue (numeric variable) per city (location variable) for a specific state. The larger the bubble representing the city, the higher the revenue for that city.

ILLUSTRATION 7.36 Choropleth Map for Tax Rates



Source: Janelle Cammenga, State and Local Sales Tax Rates, 2021, Tax Foundation. Available at https://files.taxfoundation.org/20210106094117/State-and-Local-Sales-Tax-Rates-2021.pdf

Exploration and Insights

Like the other data relationships, exploration of geospatial relationships depends on dragging and dropping. Location choices are limited to addresses, cities, states or provinces, and countries, and the location information must be formatted so that a map service, Google Maps or Bing Maps, for example, can recognize it.

On the other hand, a variety of numeric variables can be used for exploration. A map created for the HNA example could show the total revenue generated per dealership and use color intensity for encoding. That same map could display the new-to-used car sales ratio, which is an important key performance indicator (KPI) for car dealerships, and use size for encoding purposes.

Apply It 7.2
Visualize Part-To-Whole Relationships with Excel

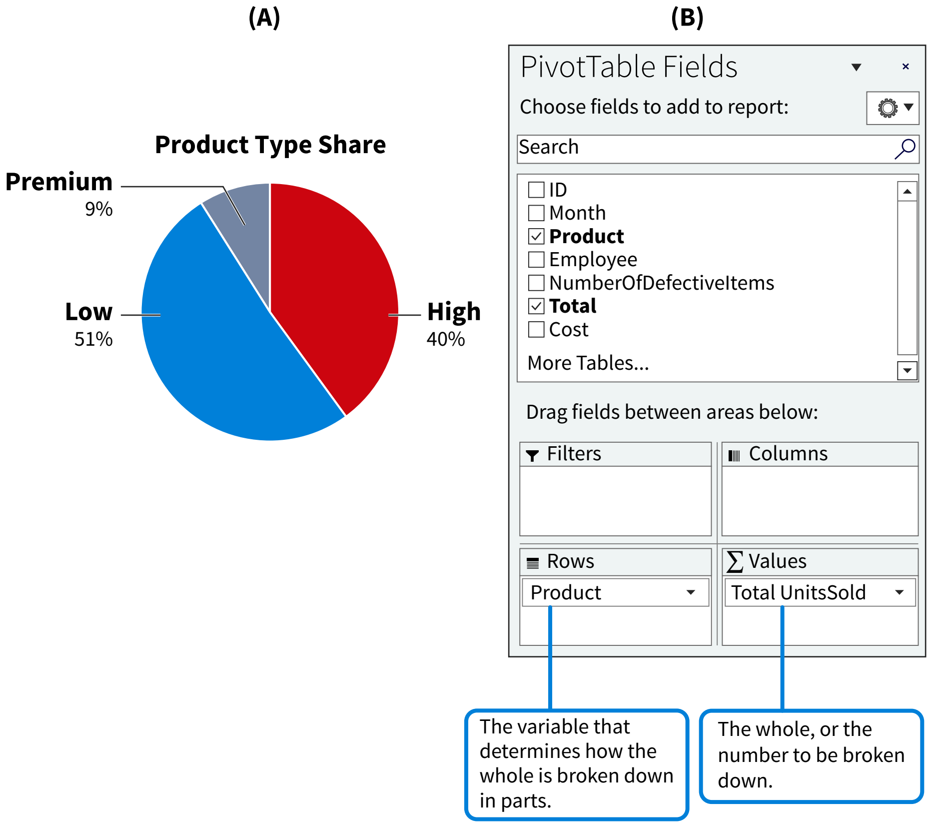
Data **Managerial Accounting** The CEO at Happy Colors needs to understand the relative importance of the company's different product types, Low, High, and Premium quality paint, in terms of the total units, or cans, being produced. Complete these three steps to perform this analysis:

1. Determine what data relationship underlies this question.
2. Identify and create a visual that could explore the data relationship.
3. Describe the insights generated by the visual.

SOLUTION

1. A part-to-whole relationship is being explored.
2. Several visualizations can be used to explore part-to-whole relationships, including pie charts, donut charts, stacked bar charts, stacked column charts, and treemap charts. The Illustration shows a pie chart (A) and the dialog box of the Excel PivotTable used for its creation (B).

Part-to-Whole Relationship with Pie Chart



3. The pie chart shows that more than half of the production is for cans of Low product type, while Premium cans have a relatively small share–lower than 10%.

7.3 How are Data Explored by Integrating Data Relationships?

LEARNING OBJECTIVE 3

Explore data by integrating foundational relationships.

While patterns one to eight explore a single, foundational data relationship with one visualization, data exploration often requires integrating two or more data relationships. Integrated data relationships can be represented with a single visualization or by using different visualizations as part of a report. Composite trends and Pareto analysis are two examples of combined data relationships that can be represented in one chart, both of which are discussed next.

Data Exploration Pattern 9: Composite Trends

A **composite trend data relationship**, or the change in a part-to-whole relationship over time, is an integrated data relationship that is often analyzed. One example is an analysis of how a business' sales mix changes over time:

- Sales mix is the relative proportion of each product in a business' total sales, which means that it is a part-to-whole relationship.
- Change over time is a time series.

We explore HNA's sales mix to illustrate this relationship, but keep in mind that there are many other business and accounting examples of composite trends, such as the changes in the structure of an investment portfolio or how the different accounts in an income statement change over time.

The exploration structure for composite trends has three variables and combines the exploration structures of the time series data relationship and part-to-whole data relationship (**Illustration 7.37**).

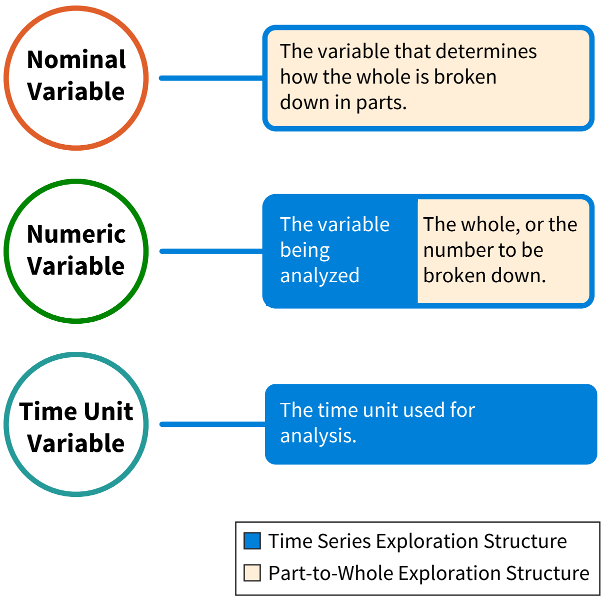


ILLUSTRATION 7.37 Exploration Structure for Composite Trend Data Relationships

Visualizations

Several visualizations, all of which rely on the exploration structure shown in Illustration 7.37, can explore composite trends. In **Illustration 7.38**, the stacked area chart is used to explore changes in HNA's unit sales mix.

ILLUSTRATION 7.38 Exploring Changes in HNA's Sales Mix with a Stacked Area Chart

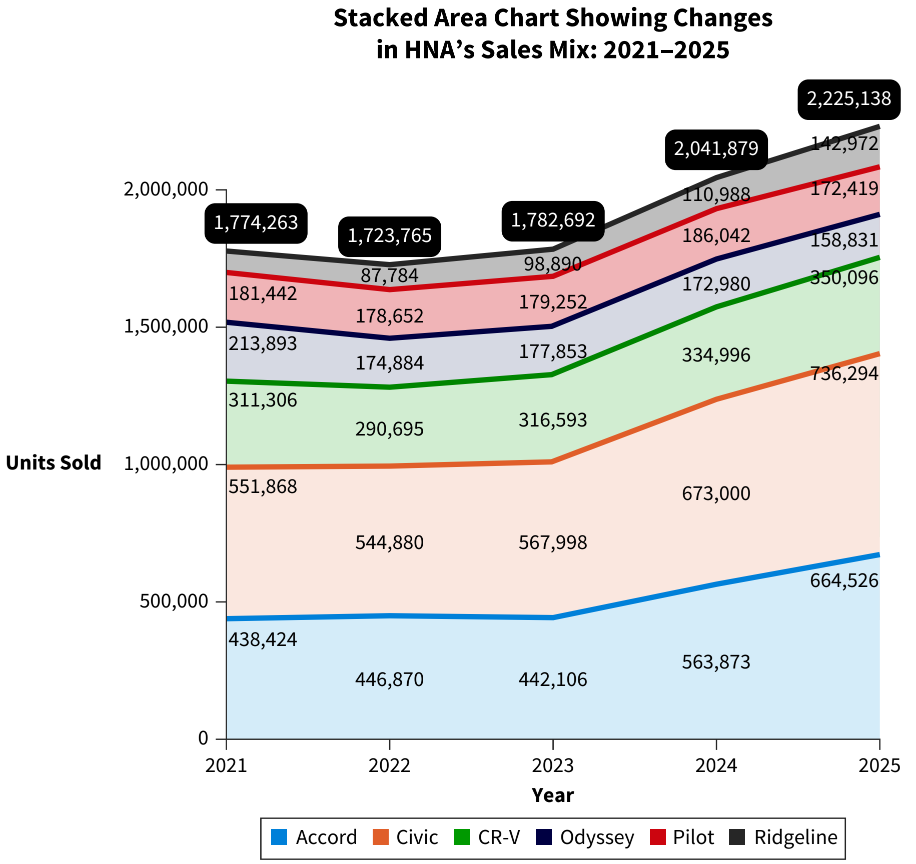
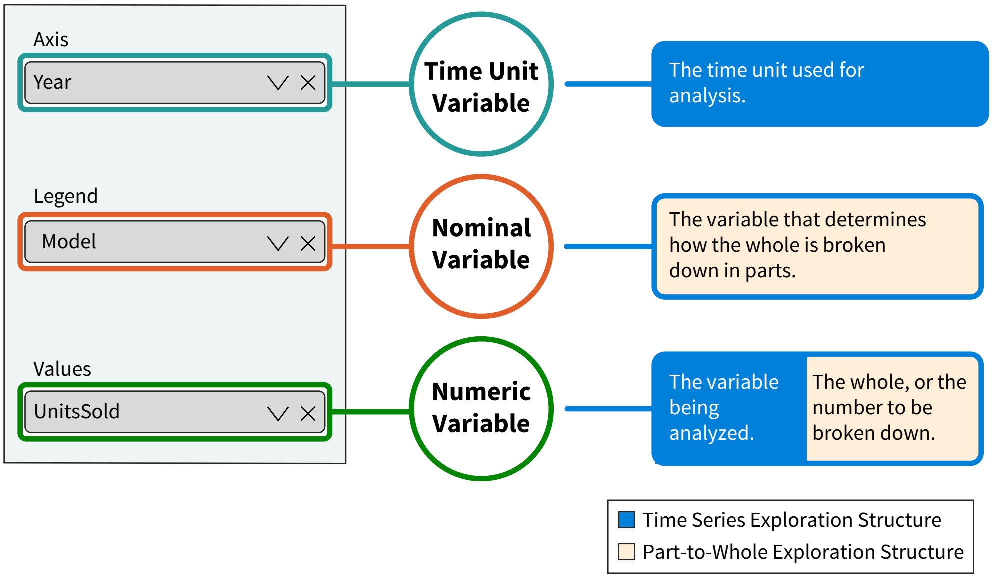


Illustration 7.39 shows how to build this chart with Power BI.

ILLUSTRATION 7.39 Power BI Definition of Stacked Area Chart



There are a few things to note about Illustration 7.38:

- The top shows how the total units sold (the numeric variable) has changed during the 2021–2025 period (the time unit variable).
- For each year, the total units sold (numeric variable) is broken down by model (the nominal variable).
- The horizontal bands reveal how the share of each model changed during the 2021–2025 period (composite trend).
- Although the data labels are helpful, the stacked columns make it challenging to understand the exact changes for most of the models.

A 100% stacked column chart (**Illustration 7.40**) using the same data is created in Power BI in the same way.

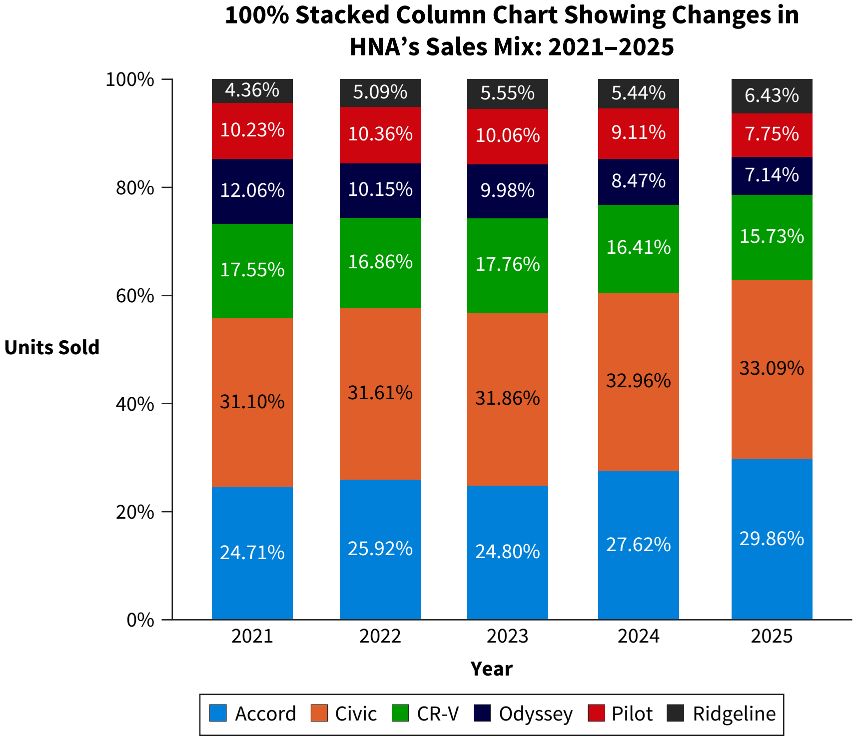


ILLUSTRATION 7.40 Exploring HNA's Sales Mix with a 100% Stacked Column Chart

While the data labels in the 100% stacked chart shown in Illustration 7.40 are more precise, this chart no longer shows the overall trend.

Exploration and Insights

Composite trends can be explored for any combination of measures (numeric variable), dimensions (nominal variable), and time units by dragging and dropping.

The stacked area chart in Illustration 7.38 integrates three insights:

- It highlights changes in the number of units sold during the 2021–2025 period (time series). The line chart in Illustration 7.32 represents the same overall trend, and both charts provide the same insight: there is an upward trend for HNA's unit sales starting from 2023.
- It reveals how much each model contributes to the annual totals (part-to-whole). In 2025, the Civic model was the best seller, followed by the Accord.
- Data labels and shaded areas indicate whether models have become more or less important in the sales mix during the 2021–2025 period (composite trend). The Accord model has grown in importance while the Odyssey model has become less important.

Compare these to the insights from the 100% stacked column chart:

- It does not show an overall trend.
- It shows the exact share each model contributes to the annual totals (part-to-whole). 33.09% of the cars sold in 2025 were Civics, and 29.86% of the cars sold in the same year were Accords.

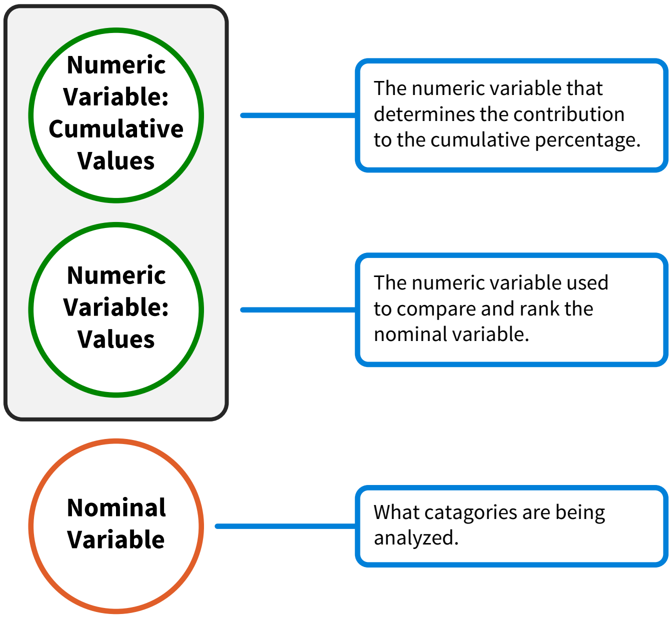
- Expressed as percentages, it reveals how a model's share in the sales mix has changed during the 2021–2025 period. The share of the Accord model grew from 31.1% in 2021 to 33.09% in 2025.

Applying the composite trend pattern can also provide insights for a business exploring how the contribution of its different regions (nominal variable) to its revenue (numeric variable) has changed during the last five years (time unit variable). A composite trend pattern can also show how the relative importance of the different accounts in an income statement changed in a specific period. This is a combination of vertical (part-to-whole) and horizontal (time series) financial statement analysis.

Data Exploration Pattern 10: Pareto Analysis

Pareto analysis (**Illustration 7.41**) determines the importance of different categories, which is nominal comparison, ranks them, and shows how each, based on that ranking, contributes to the cumulative percentage (part-to-whole).

ILLUSTRATION 7.41 Exploration Structure for Pareto Analysis



Visualizations

How To

A Pareto chart visualizes the combination of the relationships shown in the exploration structure. Some tools, like Excel, offer Pareto charts. They can also be created as line and column charts, which are supported by most tools. (**Data** How To 7.2. shows how to create a Pareto chart using both Power BI and Excel.)

Here, a line and column chart was created using Excel. **Illustration 7.42** (A) shows a PivotTable created from the HNA Unit Sales data set. It contains the different models, units sold per model, and cumulative units sold per model. Models are ranked in descending order (ranking) based on units sold. In terms of the exploration model:

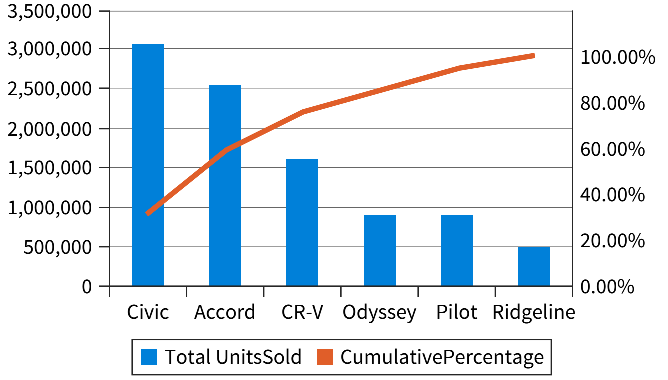
- Model represents the nominal variable.
- Total units sold represents the numeric variable (values).
- Cumulative percentage represents the numeric value (cumulative values).

Illustration 7.42 (B) is the line and column chart created from the PivotTable. Along the x-axis are the different models ranked based on total units sold. They are the items being analyzed. The line indicates the cumulative number of units sold. A second y-axis was added to emphasize that the line is expressed as a percentage. A point on the line shows how the models to the left of it cumulatively contribute to the total unit sales, as a percentage.

ILLUSTRATION 7.42 Exploring HNA's Unit Sales with Pareto Analysis

Model	TotalUnitsSold	CumulativePercentage
Civic	3,074,040	32.20%
Accord	2,555,799	58.97%
CR-V	1,603,686	75.76%
Odyssey	898,441	85.17%
Pilot	897,807	94.58%
Ridgeline	517,964	100.00%
Total	**9,547,737**	

(A) PivotTable



(B): Line and Column Chart

Exploration and Insights

The Pareto chart helps determine if there is a small group of models generating most sales. In this case, the Accord, Civic, and CR-V models together generate more than 75% of the sales, while the other three models generate less than 25%.

Pareto analysis has a range of applications in business and accounting:

- Identifying the most significant customer complaints. If a small group of issues causes most complaints, removing these issues could positively impact customer satisfaction.
- Indicating main cost categories for a specific project.
- Identifying the employees that generate the most new clients.
- Illustrating which group of holdings in a portfolio is responsible for most of its growth.

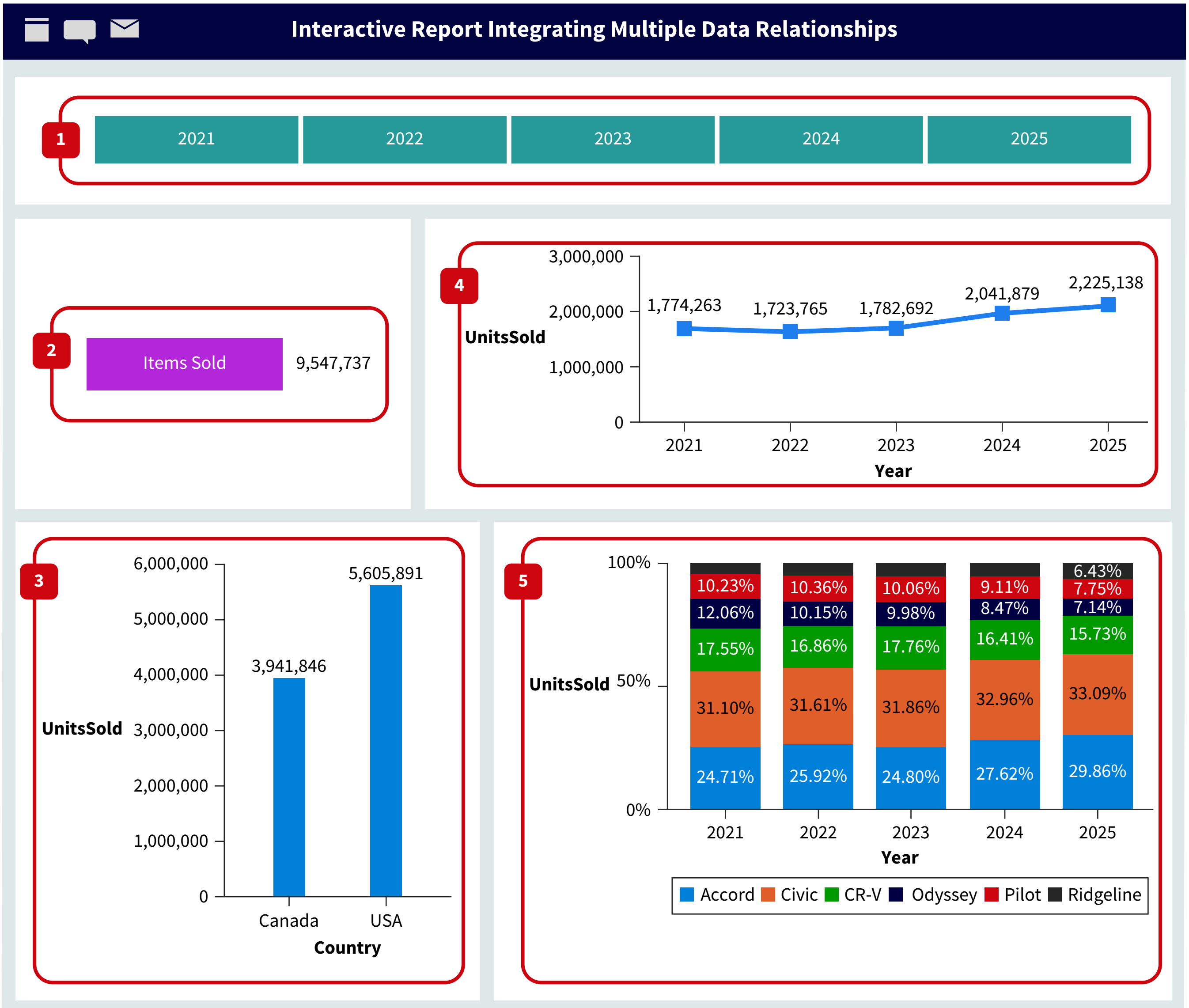
Reports Using Multiple Visualizations

Data relationships are often integrated using multiple visualizations as part of a report. Mostly interactive, these visualizations provide endless exploration opportunities. A report that combines part-to-whole and time series relationships explores trends for not only the whole, but also for each of the individual parts. A more advanced example, with multiple interactive visualizations representing different relationships, can illustrate this.

Visualizations

Illustration 7.43 is an HNA Interactive Report, which integrates data relationships with multiple visualizations.

ILLUSTRATION 7.43 Interactive Report Integrating Multiple Data Relationships



The report in Illustration 7.43 integrates five elements:

1. A slicer to select one or more years, which operates as a filter.[6]
2. A card showing the total number of units sold.
3. A column chart showing a nominal comparison—countries are compared by the total number of units sold.
4. A line chart representing a time series that reveals how the total number of units sold changed during the 2021–2025 period.
5. A 100% stacked column chart that shows composite trends.

Additionally, the report shows three different part-to-whole relationships. First, the column chart (3) breaks down the total number of units (2) by country. Second, the line chart (4) breaks down the total number of units (2) by year. Finally, the 100% stacked column chart reveals how models break down each total (per year) as percentages.

Since the report is interactive, different variables can be used for filtering purposes. The total units sold (2) can be filtered by year (1), country (3), and model (5). Also, the trendline (4) can be filtered by country (3) and model (5).

Illustration 7.44 is the exploration structure of the report in Illustration 7.43. Note that fields in one visualization, such as countries represented as bars in the column chart (3), can

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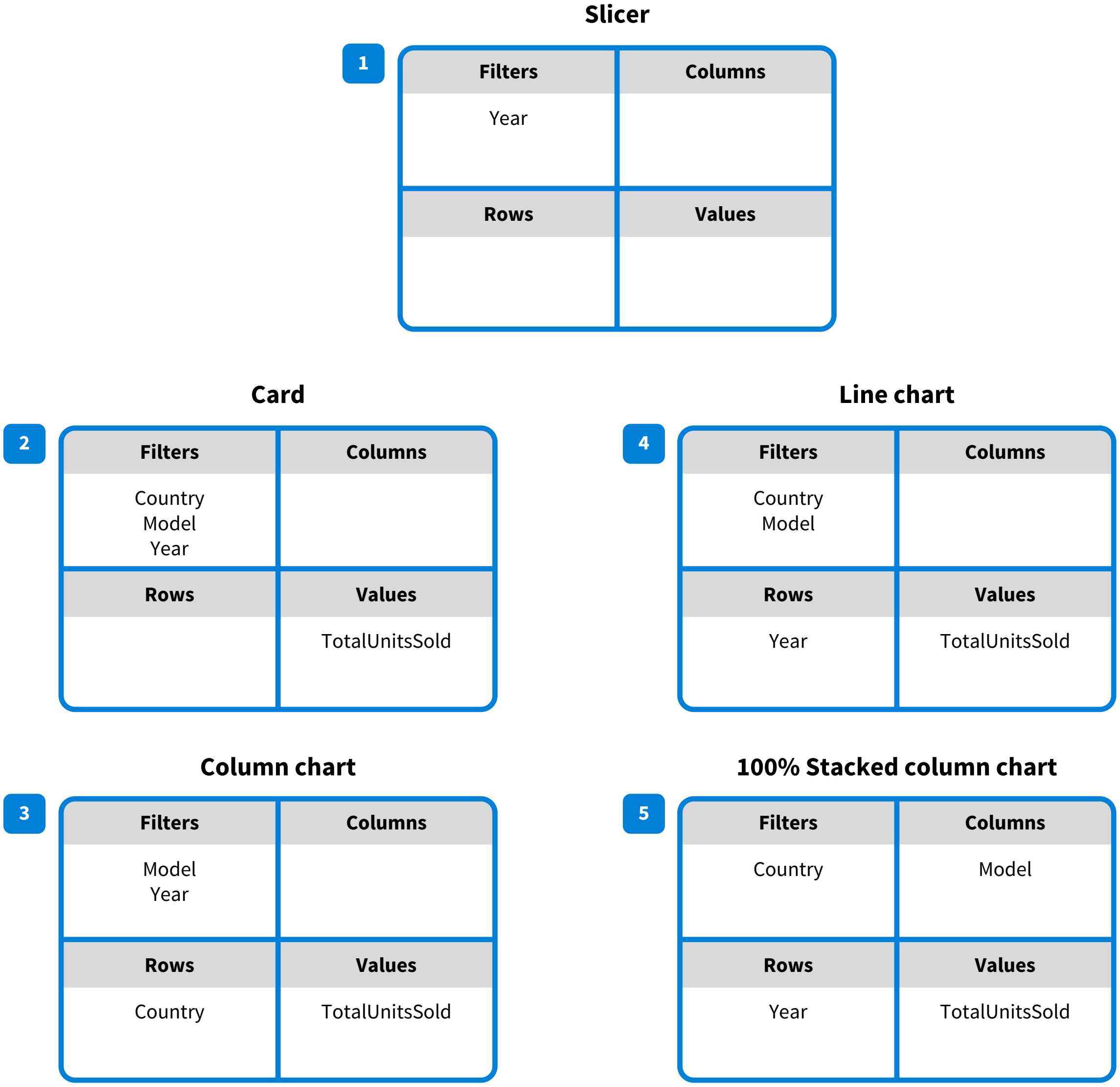
[6]We wanted to show the use of slicers as part of a report. However, this slicer is redundant since the years in the line chart and the 100% stacked column charts can be used for the same purpose.

be used as filters for the other visualizations. How this structure is supported might differ across different software tools.

ILLUSTRATION 7.44 Exploration Structure for an Interactive Report that Integrates Multiple Relationships



Exploration and Insights

What follows is an overview of the data relationships that can be explored, the insights that can be generated, and questions that can be answered with the exploration structure shown in Illustration 7.44:

- As shown in the upper-left corner of Illustration 7.44, the value shown by the card (2)—TotalUnitsSold—can be filtered by any combination of Country, Model, and Year. One example of this is how many units were sold in the USA (country) of the Pilot model (model) in 2022 (year).
- The column chart (3) compares the units sold (TotalUnitsSold) across countries. Using Model and Year as filters (from the other visualizations), makes it possible to compare Canadian and USA sales of the Accord and Civic models in 2025.
- The line chart (4) analyzes the overall sales trend. However, trends could also be shown for specific combinations of country and model, such as exploring trends for Ridgeline sales in Canada.
- The 100% stacked column chart (5) determines the units sold per model per year and presents them as a percentage.

Once again, using filters allows exploring more specific scenarios, such as selecting USA in the column chart (3) and 2025 in the line chart (4).

Illustration 7.45 shows how the report would look after using these filters.

ILLUSTRATION 7.45 U.S. 2025 Sales Mix

Data exploration patterns are best practices for performing analysis. They make us more aware of data relationships, how they can be represented and explored, the insights that can be generated, and how they can be integrated.

Apply It 7.3
Build an Interactive Report

Data **Managerial Accounting** Defective paint cans are an issue for Mika at Happy Colors, so she would like to explore the following:

- Compare the overall number of defective units (paint cans) per employee.
- Determine how the relative proportion of defective units across products differs among employees.
- Compare the relative number of defective units per employee.

Implement the following four steps to help perform the analysis:

1. Determine what data relationships underlie this analysis.
2. Identify the visualizations that could be used to explore these data relationships.
3. Build an interactive report with these visualizations and draw the report's exploration structure.
4. Identify the insights generated by your analysis.

SOLUTION

1. The data relationships underlying this analysis:
   - Nominal comparison: Compare the overall number of defective units per employee.
   - Part-to-whole: Per employee, relative proportion of defective units across products.
   - Nominal comparison: Compare the relative number of defective units per employee.
2. Different visualizations can be selected for these data relationships. We will use column charts to explore nominal comparison and a 100% stacked column chart to explore the part-to-whole relationship. For further exploration with different scenarios, we are adding month and product filters.
3. There are many ways to conduct this analysis. The following is a possible interactive report implemented with Power BI.

The exploration structure for this report looks as follows:

Exploration Space: Happy Colors Defective Cans Analysis

1 Slicers

Filters	Columns
Month Product	
Rows	**Values**

3 100% Stacked column chart

Filters	Columns
Month	Product
Rows	**Values**
Employee	Total Number of Defective Cans

2 Column chart

Filters	Columns
Month Product	
Rows	**Values**
Employee	Total Number of Defective Cans

4 Column chart

Filters	Columns
Month Product	
Rows	**Values**
Employee	Relative Number of Defective Cans

4. These insights can be collected from the report:

- The column chart (2) can compare the number of defective cans per employee. This is nominal comparison. Andre has the highest number and Camille the lowest.
- The 100% stacked column chart (3) breaks down the number of defective cans by employee and product. This is a part-to-whole relationship. Most of Bianca's defective cans are for Low products. For Camille, most of the defective cans are for High products.
- The second column (4) chart mirrors the first column chart (2) and is also a nominal comparison. However, here, an employee's number of defective cans is shown as a percentage of the total number of defective cans (part-to-whole relationship).
- Using filters (1), select any combination of month and product, which generates additional insights. For example, in January, Andre had more defective cans than Rebecca. However, in February it was the other way around.

Chapter Review and Practice

Learning Objectives Review

1 Describe the process of data exploration.

Data exploration is the discovery process of looking for something new and previously unknown:

- Data exploration has four steps: identifying questions, identifying data relationships, exploring those relationships, and generating insights.
- A widely used tool for data exploration is the Excel PivotTable. It enables exploration by selecting, from a list of fields, the numbers to be analyzed, how the numbers will be broken down in terms of rows and columns, and the filters that apply to the data.
- PivotTables integrate the key techniques of data exploration: identifying data relationships, filtering, and dragging and dropping.

2 Explore foundational data relationships through visualizations.

Investigating data relationships is at the heart of data exploration. Eight foundational patterns can help answer questions about which data relationships can be explored, how they can be illustrated in visualizations, how to create those visualizations, and what insights can be generated from them:

- A nominal comparison data relationship compares the values of a nominal variable based on the value of a second, numeric, variable. For example, we can compare product categories (nominal variable) based on the total profit they have generated (numeric variable).
- A distribution data relationship shows how the values of a quantitative variable are spread out. This relationship can show the highest price paid for a product, the lowest price, the price range, etc.
- A deviation data relationship shows how a set of actual values deviate from their reference values. Exploring a deviation relationship can show how the actual profit for 2025 compares with the budgeted profit.
- A ranking data relationship orders the values of a variable sequentially based on the values of a second variable. An example of what it can show is how different products rank based on profit.
- A part-to-whole data relationship displays how each part compares to the whole and how the parts compare to each other. It helps to understand issues such as the relative contribution of each product category to a business' total revenue.
- A correlation data relationship indicates the degree to which two variables move in the same or opposite direction. For example, how ice cream sales are correlated with temperature.
- A time series data relationship defines the values of a variable at sequential points in time. For example, it can be used to explore how a business' profit changed during the last decade.
- A geospatial data relationship encodes locations through coloring, shades, and size, based on the values of a numeric variable. States (location) can be colored based on the total profit (numeric variable) they have generated.

3 Explore data by integrating foundational relationships.

Data exploration often requires the integration of two or more data relationships, which can be represented by a single visualization or by using different visualizations as part of a report:

- A composite trend is an example of integrated data relationships that portray how relative proportions (part-whole) change over time (time series). Exploring these relationships can help answer questions, such as how did a business' sales mix fluctuate the last ten year.
- Pareto analysis ranks categories and determines, based on that ranking, how much the categories contribute to the cumulative percentage. Pareto charts can explore data to determine, for example, the most frequent surgical errors in a hospital.
- Interactive reports with multiple visualizations that represent different data relationships provide endless exploration opportunities. A business' profit total (visual 1) can be broken down by product category (visual 2) while product categories can be further broken down by region (visual 3) or the other way around.

Illustration 7.46 is an overview of the ten data exploration patterns presented in this chapter.

ILLUSTRATION 7.46 Overview of Data Exploration Patterns

Id	Data Relationship	Description	Visualization Examples (Bold indicates chapter example)
Foundational Data Relationships			
1	Nominal Comparison	Compares the values of a nominal variable based on the values of a second, numeric variable.	Bar chart, **column chart**, dot plot, lollipop chart
2	Distribution	Shows how the values of a numeric variable are distributed, or spread.	Histogram, **box-and-whisker (boxplot) chart**, violin plot
3	Deviation	Shows how a set of actual values deviate from their reference values such as budgeted or forecasted values.	**Clustered bar** and column **charts**, gauge, bullet chart
4	Ranking	Orders the values of a variable sequentially based on the values of a second variable.	Defined as part of a table, bar chart, or **column chart**.
5	Part-To-Whole	Shows how each part compares to the whole and how the parts compare to one another.	**Pie chart**, donut chart, stacked bar chart, stacked column chart, treemap chart
6	Correlation	Indicates the degree to which two variables move in the same or the opposite direction.	**Scatterplot**
7	Time Series	Shows the values of a variable at sequential points in time.	Area chart, bar chart, column chart, **line chart**, sparkline chart, waterfall chart
8	Geospatial	Assigns numeric values to locations and encodes them through coloring (shade) and size (bubble size).	**Maps**
Integrated Data Relationships			
9	Composite Trends	Shows how a composite structure (part-to-whole) changes over time (time series).	Stacked column chart, **100% stacked column chart, stacked area chart**
10	Pareto Analysis	Illustrates the importance of different categories (nominal comparison), ranks them (ranking), and shows their cumulative percentage (part-to-whole).	Pareto chart, **line and column chart**

Key Terms Review

Composite trend data relationship 7-27
Correlation data relationship 7-20
Data exploration 7-1
Data relationship 7-3
Deviation data relationship 7-15
Distribution data relationship 7-12
Exploration structure 7-10
Geospatial data relationship 7-24
Insight 7-2
Nominal comparison data relationship 7-10
Pareto analysis 7-30
Part–to-whole data relationship 7-18
Ranking data relationship 7-17
Time series data relationship 7-22

How To Walk-Throughs

How To

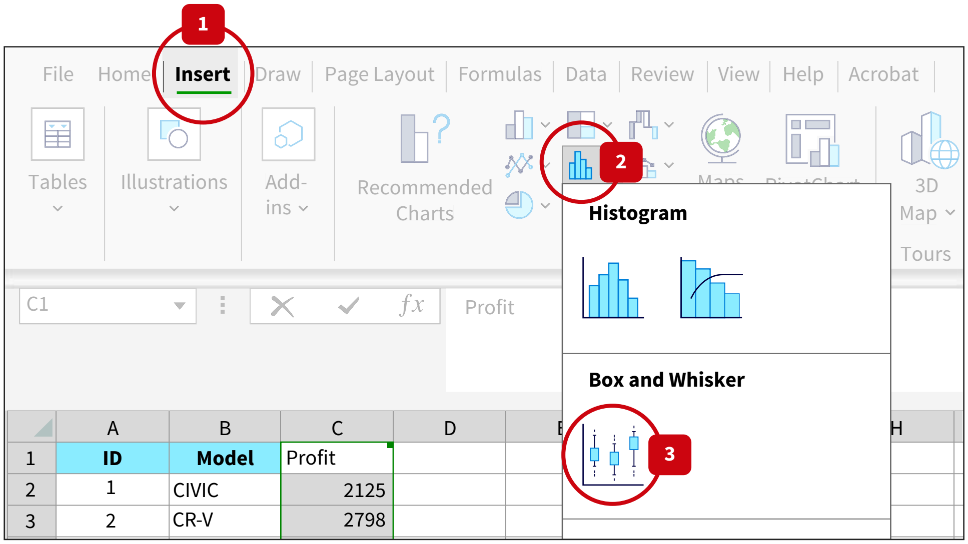
HOW TO 7.1 Create a Box-and-Whisker Chart with Excel

Box-and-whisker charts are excellent tools for exploring distribution relationships. Illustration 7.14 showed an example of a box-and-whisker chart created with Power BI, but you can also create one with Excel.

What You Need: **Data** The How To 7.1 data file.

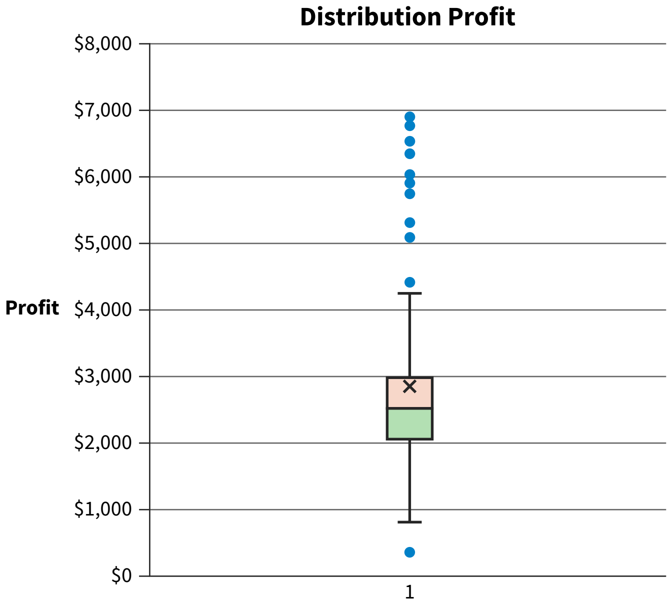
STEP 1: Select the data in the **Profit** column as shown in **Illustration 7.47**, then click the **Insert** tab (1). Go to the **Charts** group and select **Insert Statistic Chart** (2) and **Box-and-Whisker** (3).

ILLUSTRATION 7.47 Creating a Box-and-Whisker Chart with Excel



This creates the box-and-whisker chart shown next (**Illustration 7.48**).

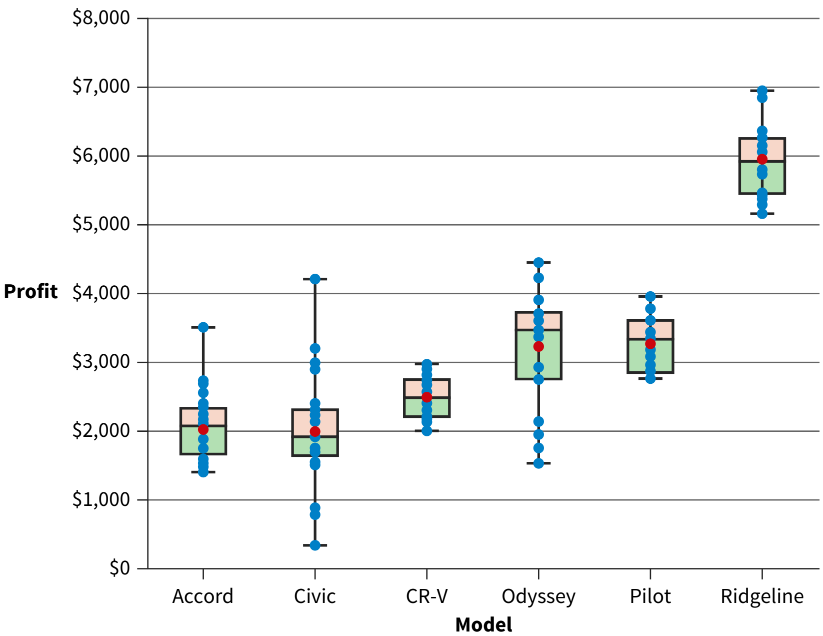
ILLUSTRATION 7.48 Excel Box-and-Whisker Chart for Profit Distribution



The X symbol represents the average. Outliers, defined as being 1.5 times of the interquartile range (IQR) below the first quartile or above the third quartile, are also shown as part of the chart. To see all the data points, click one of the boxes and select **Format Data Series**. Then, select **Show Inner Points**. We applied some additional formatting to the chart.

STEP 2: To create parallel profit distributions for the different models, select both the **Model** and **Profit** columns. To recreate the chart in **Illustration 7.49**, sort the **Model** column in ascending order.

ILLUSTRATION 7.49 Parallel Profit Distributions for HNA Car Models.



How To

HOW TO 7.2 Create a Pareto Chart with Excel and Power BI

When conducting a Pareto analysis, you must calculate the data, create the chart, and perform the analysis. This can be done in Excel or in Power BI. Here, we illustrate how to combine both tools.

What You Need: Data The How To 7.2 data file.

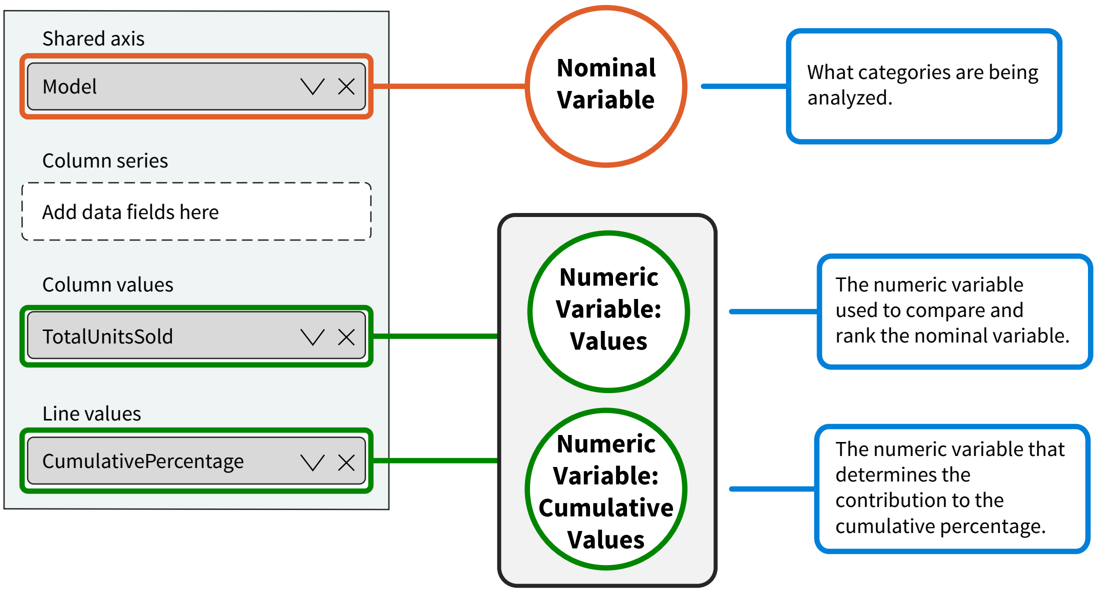
STEP 1: Open Power BI and extract the Pareto raw data worksheet from the HNA Pareto raw data Excel file. The worksheet is shown in **Illustration 7.50** and is a copy of the PivotTable data shown in Illustration 7.42 (A).

ILLUSTRATION 7.50 Pareto Data Worksheet

Model	TotalUnitsSold	CumulativePercentage
Civic	3,074,040	32.20%
Accord	2,555,799	58.97%
CR-V	1,603,686	75.76%
Odyssey	898,441	85.17%
Pilot	897,807	94.58%
Ridgeline	517,964	100.00%

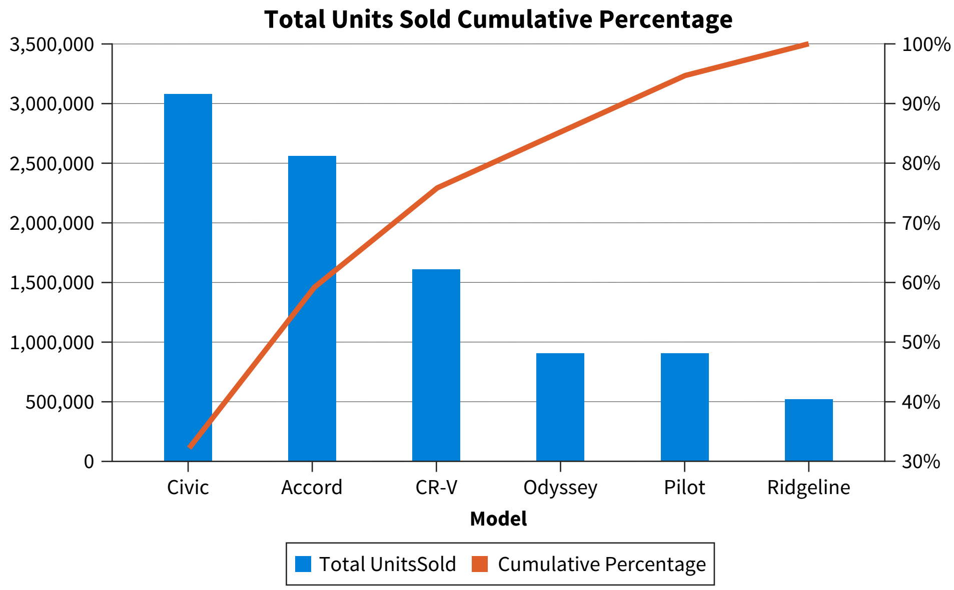
STEP 2: Select the Line and Stacked Column chart in the **Visualizations** pane. The definition for the Pareto chart is shown in **Illustration 7.51**.

ILLUSTRATION 7.51 Creating a Line and Column Chart for Pareto Analysis in Power BI



The corresponding chart is shown in **Illustration 7.52** and is similar to the chart in Illustration 7.42 (B). Some additional formatting was applied.

ILLUSTRATION 7.52 Pareto Analysis of HNA's Unit Sales



Combining multiple tools for a data analytics task and integrating their strengths is common practice.

Data The Data tag appears when the data required to answer a question or complete an exercise are available on Wiley's online learning platform.

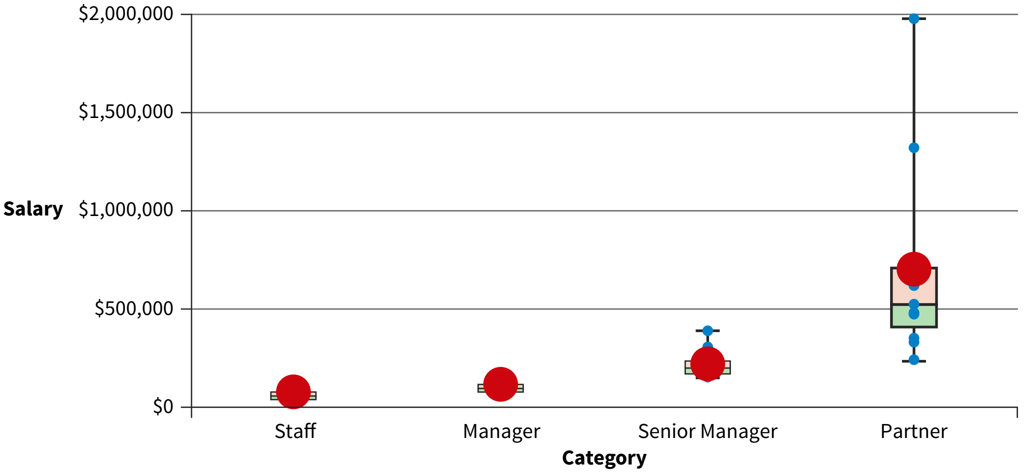
Multiple Choice Questions

1. **(LO 1)** Exploring data is a four-step process of identifying questions, identifying data relationships, exploring those relationships, and
   - **a.** identifying information requirements.
   - **b.** correcting data anomalies.
   - **c.** identifying insights.
   - **d.** communicating insights.

2. **(LO 1)** Data exploration relies on four elements. Which of the following is not one of the four elements?
   - **a.** Cells
   - **b.** Filters
   - **c.** Rows
   - **d.** Values
   - **e.** Columns

3. **(LO 1)** A PivotTable's values area
   - **a.** determines the number to be reported.
   - **b.** determines what is being analyzed.
   - **c.** represents the variables used for slicing and dicing.
   - **d.** represents data points.

4. **(LO 1)** Sweet Candies is a gourmet shop selling candy baskets to customers in all 50 U.S. states. They built a PivotTable in Excel to analyze the revenues generated by the different mid-Atlantic states in January 2025. Which of the following components of the PivotTable will be empty?
   - **a.** Columns
   - **b.** Filters
   - **c.** Rows
   - **d.** Values
5. **(LO 1)** Which of the following tools explores data?
   - **a.** Microsoft Excel
   - **b.** Microsoft Power BI
   - **c.** Tableau
   - **d.** All of these tools can explore data.
6. **(LO 2)** The following are all data relationships *except*
   - **a.** geospatial.
   - **b.** correlation.
   - **c.** part-to-whole.
   - **d.** deviation.
   - **e.** boolean.
   - **f.** distribution.
7. **(LO 2)** When comparing overtime among employees to determine who has no overtime hours, few overtime hours, or many overtime hours, you are exploring what data relationship?
   - **a.** Part-to-whole
   - **b.** Ranking
   - **c.** Nominal comparison
   - **d.** Distribution
8. **(LO 2)** Which of the following does *not* describe data exploration?
   - **a.** Creating a list of products, ordered by Facebook "likes," in descending order.
   - **b.** Creating a visualization showing, per product, the 2025 actual and targeted number of units sold.
   - **c.** Creating a chart showing the mean, median, and range of last year's sales transaction amounts.
   - **d.** Creating a map with states or provinces colored based on the total quantity sold there.
   - **e.** Creating a formula that defines gross profit: revenue – cost of goods sold.
9. **(LO 2)** Which of the following is *not* an example of critical thinking applied to data exploration?
   - **a.** Looking at data from different viewpoints, such as customers, investors, and employees.
   - **b.** Evaluating and re-evaluating the assumptions underlying the insights generated.
   - **c.** Transforming dashboards to make sure the insights generated are easy to understand.
   - **d.** Studying and applying advanced statistical methods to add precision to the insights generated.
10. **(LO 2)** The box-and-whisker chart shows salary distributions for the different categories of employees at a public accounting firm: staff, manager, senior manager, and partner.



Insights for this data set could include:

1. The salaries for partners are significantly higher than for the other categories.
2. There is little variation in the salaries for staff members.
3. The salaries for partners are skewed–there is more variation among the high salaries than among the low salaries.
4. There is an outlier for the senior manager category.

Which of these insights can be derived from the box-and-whisker chart?

a. 1 and 2
b. 1 and 3
c. 1 and 4
d. 1, 2, and 3
e. 1, 2, and 4
f. 1, 2, 3, and 4

11. **(LO 2)** Flame, a U.S. company that sells electronic products worldwide, gives you this data set.

Region	2020 Profits
North-America	19,922,189
South-America	89,900,011
Western-Europe	3,008,223
Eastern-Europe	2,347,781
Africa	890,002
Asia	122,343

What data relationship is *not* relevant for this data set?

a. Nominal comparison
b. Ranking
c. Geospatial
d. Part-to-whole
e. Deviation
f. All of these are relevant.

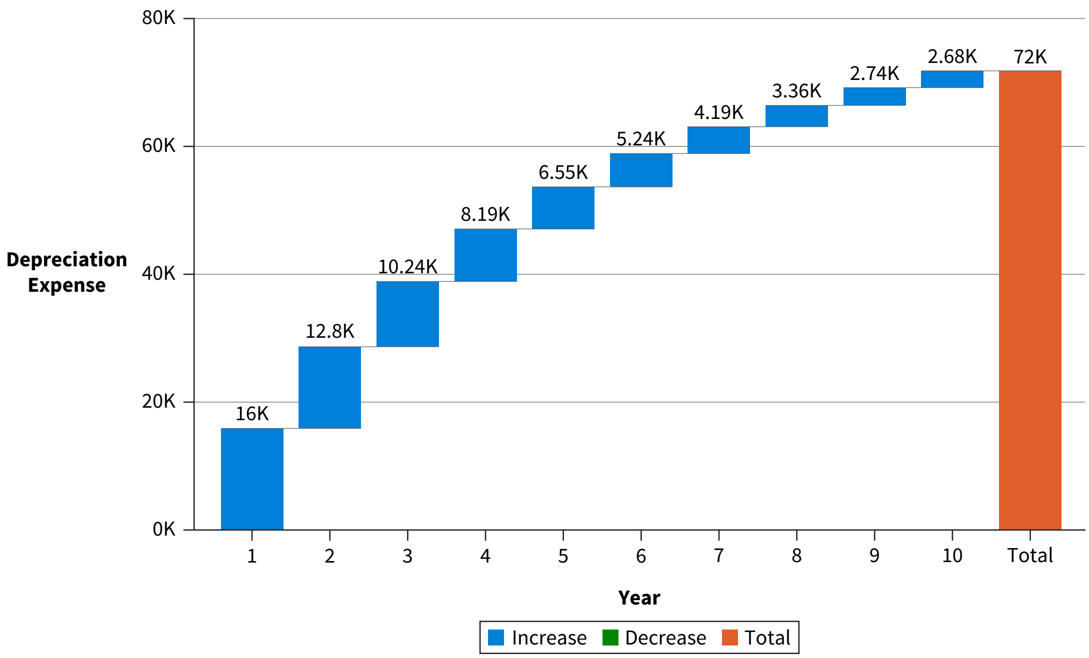
12. **(LO 2)** Which of the following aspects is *not* relevant for a time series data relationship?

a. Trend
b. Seasonal pattern
c. Magnitude of discrepancies
d. Irregularities
e. Forecasts

13. **(LO 2)** Which of the following aspects is *not* relevant for a correlation data relationship?

a. Clusters
b. Strength (correlation coefficient)
c. Direction (positive or negative)
d. Outliers
e. Forecasts

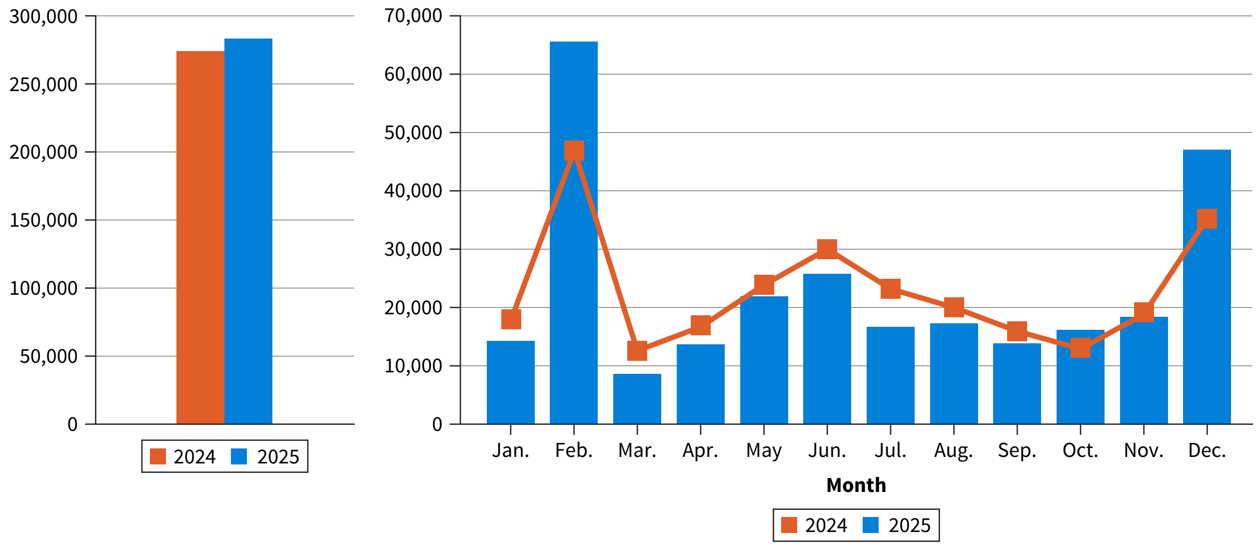
14. **(LO 3)** ABC company bought a machine with an estimated useful life of 10 years for $80,000 (cost). The residual (salvage) value of the machine is $8,000.



The waterfall chart visualizes the machine's depreciation using what data relationships?

a. Part-to-whole, time-series
b. Deviation, time-series
c. Time-series, nominal comparison
d. Ranking, time-series
e. Deviation, part-to-whole

15. **(LO 3)** Delvoux is a high-end manufacturer of handbags located in Brussels, Belgium. They created this report for exploring units sold.



Which data relationships are explored in this visualization?

- **a.** Ranking, time-series
- **b.** Nominal comparison, part-whole, ranking, time series
- **c.** Deviation, distribution, time series
- **d.** Deviation, nominal comparison, part-whole, time series

16. **(LO 3)** Cone Canoes (CoCaN) is a high-quality canoe manufacturer that sells their products worldwide. They make four different types of canoes: recreational, whitewater, racing, and fishing. They would like to further explore the total revenues generated by U.S. sales of whitewater and racing canoes for all four quarters in 2025. Through their ERP, they have access to their sales data for the last ten years. Which of the following represents the exploration structure for such an analysis?

A

Filters	Columns
Country ProductType	Year
Rows	**Values**
ProductType	TotalRevenues

B

Filters	Columns
Country Year ProductType	Quarters
Rows	**Values**
ProductType	TotalRevenues

C

Filters	Columns
ProductType Year	
Rows	**Values**
Country Year	TotalRevenues

D

Filters	Columns
Country Year	Quarters
Rows	**Values**
ProductType	TotalRevenues

17. **(LO 3)** Vosci Salon de Spa, a hair salon in Albuquerque, New Mexico offers three service lines: hair, nails, and tanning. Vosci's owner provides you with the following summary:

	Hair	Nails	Tanning	Totals
Revenues	$ 123,451	$ 37,987	$ 55,231	$ 216,669
Variable Costs	$ 79,862	$ 21,324	$ 13,412	$ 114,598
Fixed Costs	$ 18,776	$ 15,112	$ 25,771	$ 59,659
Profits	$ 24,813	$ 1,551	$ 16,048	$ 42,412

Which of the following statements is incorrect?

a. There is a part-to-whole relationship between revenues (whole) and variable cost, fixed costs, and profits (parts).

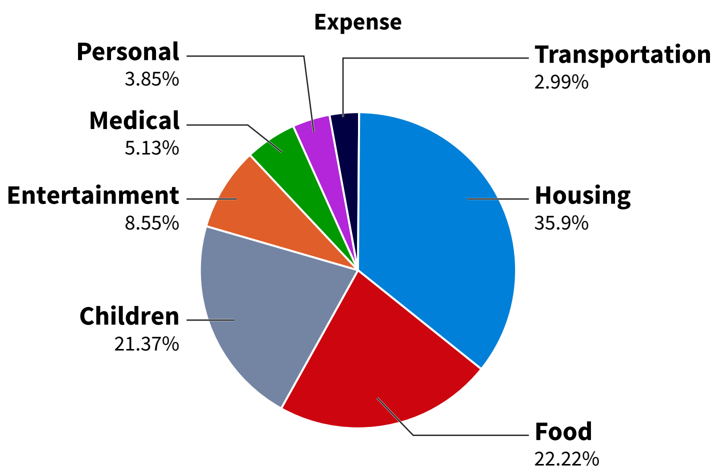
b. Profit margin is the relative proportion of profits as part of revenues and thus based on a part-to-whole relationship.

c. Each row in the table presents a time series which enables to compare the relative importance of different services.

d. The different service lines can be compared based on profits (nominal comparison).

Review Questions

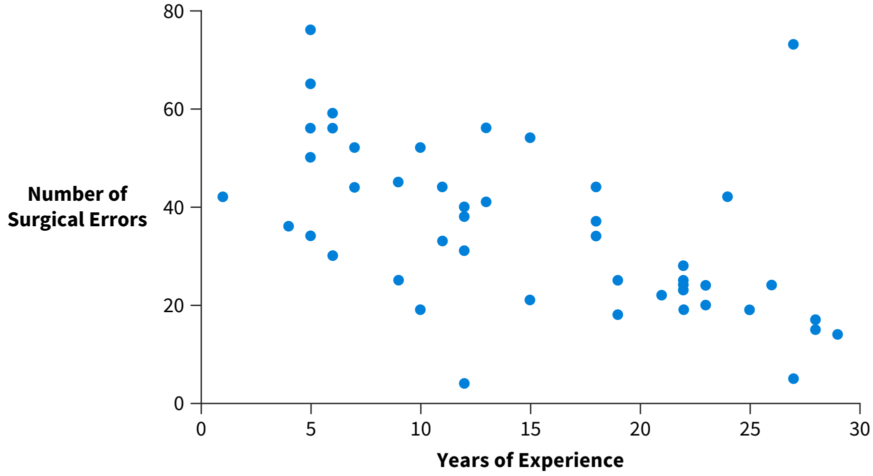
1. **(LO 1)** Describe the goal of data exploration.
2. **(LO 1)** What are the four steps in the data exploration process?
3. **(LO 1)** How would you describe an insight in the context of data exploration?
4. **(LO 1)** Briefly describe the role of charts in data exploration.
5. **(LO 1)** What is a PivotTable's key strength?
6. **(LO 1)** What three key data exploration techniques are supported by PivotTables?
7. **(LO 1)** Discuss the purpose of the Values area in a PivotTable.
8. **(LO 1)** What is the purpose of filters in data exploration?
9. **(LO 2)** What are the two key variables in a nominal comparison relationship? Use an example to explain.
10. **(LO 2)** Discuss two insights that can be generated from exploring a distribution relationship. Use a corporation's earnings during the last twenty years as the example for your discussion.
11. **(LO 2)** List at least five different pieces of information that a box-and-whisker chart provides about a distribution.
12. **(LO 2)** What role does a nominal variable play when creating a box-and-whisker chart?
13. **(LO 2)** A university compares its faculty salaries with peer institutions (peer salary comparison). What kind of relationship are they exploring? What kind of insights might they generate and what decisions can be affected by such insights?
14. **(LO 2)** The pie chart summarizes the expenses of a household.



1. What data relationship is being explored?

2. What are two insights that can be generated from the pie chart?

15. **(LO 2)** Beebe is a hospital in southern Delaware that employees fifty surgeons. For insurance purposes they must provide detailed data regarding surgical errors. One of the reports they prepare is shown here.

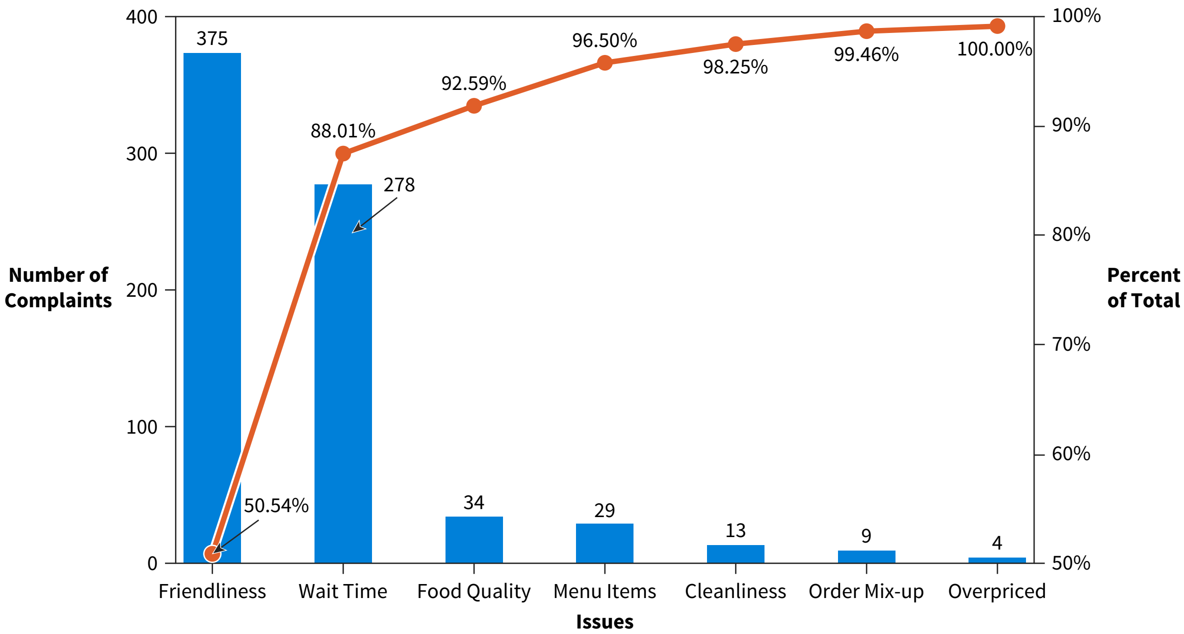


1. What chart is being used?
2. What data relationship is being presented?
3. What two insights regarding the data relationship can you identify?

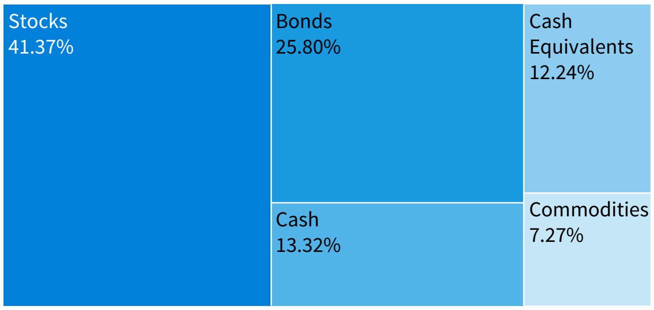
16. **(LO 2)** What are the two mechanisms that can be used to encode locations?

17. **(LO 3)** Analysis of composite trends is based on three variables: a nominal variable, a numeric variable, and a time unit variable. Discuss the role of each of these variables in the analysis. Use the medal count per country during the last five summer Olympics as an example for your discussion.

18. **(LO 3)** Aurora is a pizza restaurant in Alaska. Recently, business has been significantly slower than normal for them. To better understand why this is happening, they created a Pareto chart using data from the complaints they received on Yelp. A Pareto chart combines several data relationships. Name these relationships and discuss them in the context of the chart shown.



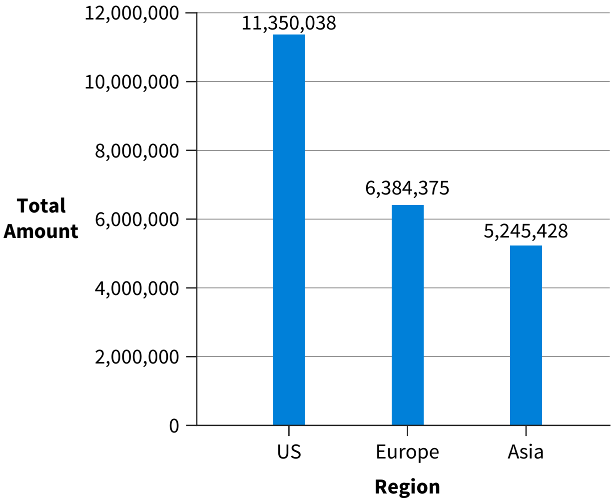
19. **(LO 3)** What are the two relationships integrated by a treemap? Explain how the following tree map shows these relationships.

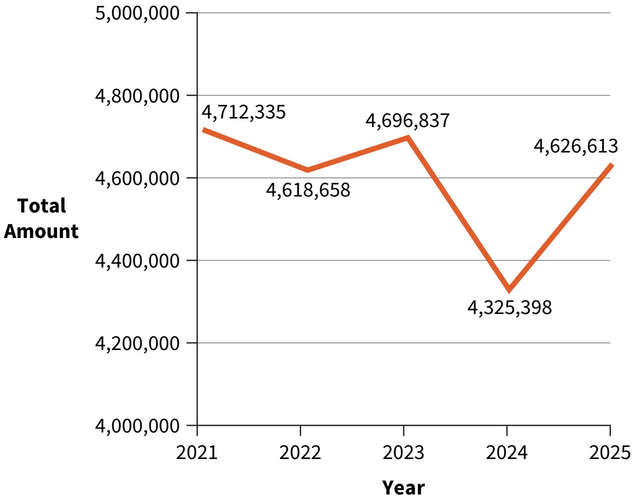


20. **(LO 3)** ESpin is a company in Surprise, AZ, manufacturing revolutionary toothbrushes that provide deeper cleaning than average toothbrushes. They give you data for their 2021–2025 sales in their three current regions. The data is shown here.

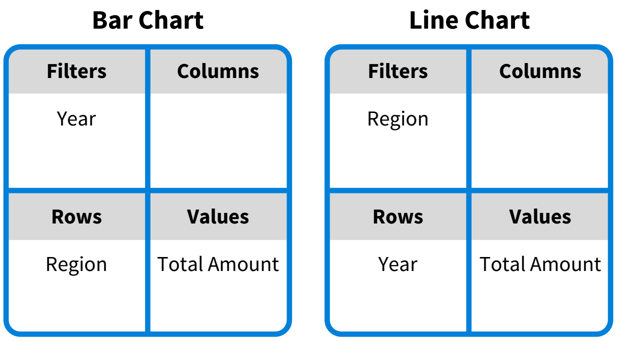
Region	2021	2022	2023	2024	2025
Asia	789,112	977,111	1,009,112	1,125,521	1,344,572
Europe	1,256,111	1,238,771	1,398,814	1,199,765	1,290,914
U.S.	2,667,112	2,402,776	2,288,911	2,000,112	1,991,127

You develop the following interactive report for them.





The corresponding exploration structure follows:



Determine at least three questions that can be answered with this report.

Brief Exercises

BE 7.1 **(LO 1)** Match the appropriate term to each definition. Terms can be used once, more than once, or not at all.

a. Outlier
b. Correlation
c. Insight
d. Analytical database
e. Drag and drop
f. Data interpretation
g. Data exploration

Statement	Term
1. An observation that might significantly affect a business' decision-making.	
2. Input of data exploration.	
3. Output of data exploration.	
4. The discovery process of learning something new and previously unknown from data.	

BE 7.2 **(LO 1)** Data exploration is based on three elements: (1) the number being analyzed, (2) variables that break down the number, and (3) ________________.

BE 7.3 **(LO 1)** **Data** **Financial Accounting** Mario runs a Food Truck, the Treat Buggy, in Atlantic City, NJ. Use the data file provided by Mario to determine the percentage of his revenues during summer months (June, July, August) generated from ice cream sales compared to sales from other items.

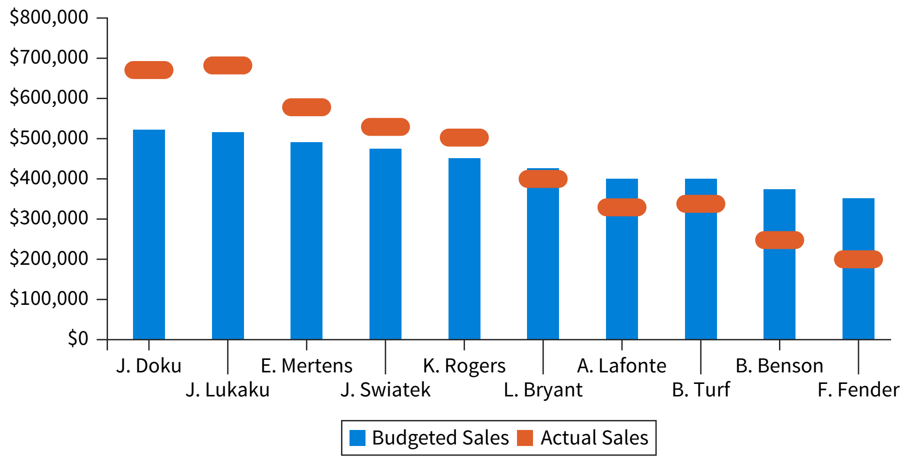
BE 7.4 **(LO 1)** **Data** **Managerial Accounting** Mauretius is a boutique Monegask car builder. They manufacture 5,000 cars each year. There are two different car models: Perfic and Fortis. A critical component for all their cars is a high-performance battery. The company outsources the manufacturing of the batteries for both models to three vendors: Optima, Champion, and Diehard. Specifications and prices are the same for all three vendors. Because of this, vendors are rated based on reliability. They are evaluated as to the extent they comply with Mauretius' specifications. Batteries that are not compliant are returned. Using the data set provided by Mauretius, compare vendors' reliability as a percentage for both models. Based on reliability, who should be the preferred vendor for each battery type?

BE 7.5 **(LO 2)** **Managerial Accounting** Wallmark is a fast-growing manufacturer of musical greeting cards in Tallahassee, FL. To improve their business and manage growth they are heavily investing in data analytics. Match each of the data exploration scenarios with one of the following data relationships.

a. Correlation
b. Deviation
c. Distribution
d. Part-to-whole

	Data Exploration	Data Relationship
1.	A strong growth in Wallmark's number of employees has resulted in strong growth in office expenses.	
2.	Last year Wallmark's labor costs systematically exceeded their estimates. This might have been caused by the premiums Wallmark paid related to the pandemic.	
3.	Last month the average price Wallmark paid for a button battery, a key element in musical greeting cards, was $7. However, prices for the exact same product ranged from $3 to $18.	
4.	Batteries represent 20% of the cost of making a standard musical greeting card.	

BE 7.6 **(LO 2)** **Managerial Accounting** Monar is a trendy shoe store on Fifth Avenue in New York that employs ten in-store salespeople. Monar's CEO uses the following chart when analyzing the performance of the company's salespeople. The chart shows salespeople in descending order of seniority. Freddy Fender is the most recent hire.



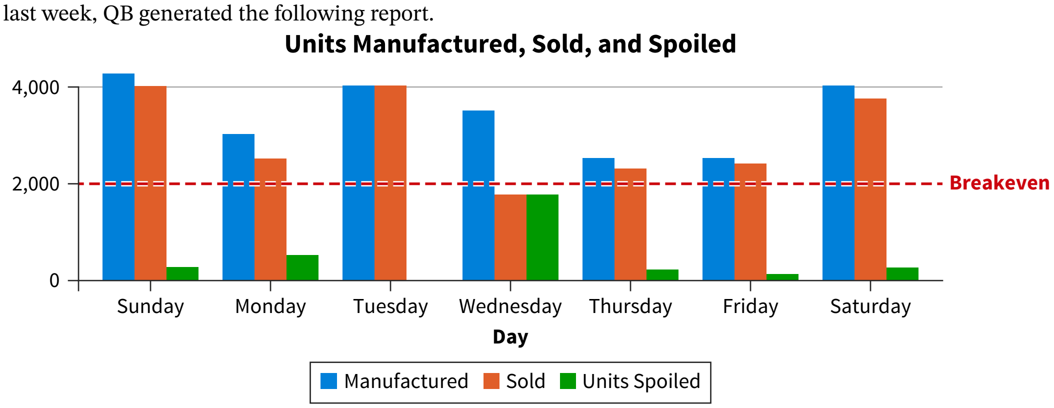
1. What data relationship is being explored?
2. What insights can be generated from this chart?

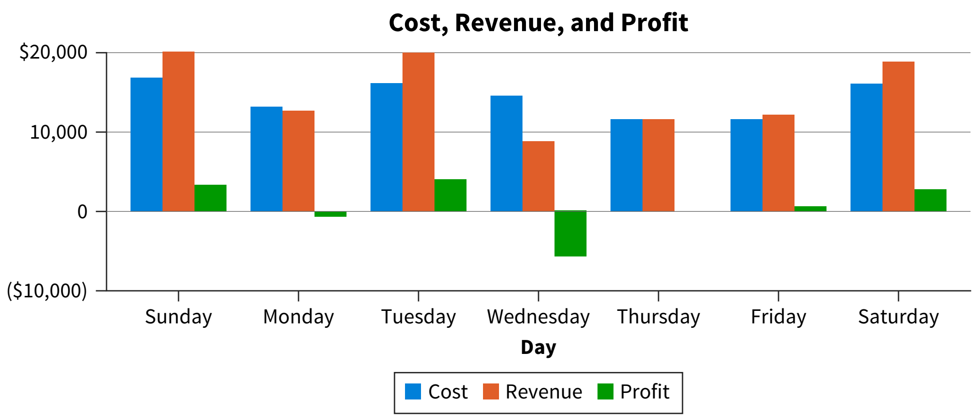
BE 7.7 (LO 2) Data Managerial Accounting Raisin is an electronics manufacturer with headquarters in Dijon, France. Its CEO wants a dashboard that shows the progress made for each product line compared to last year's unit sales. The goal is a 10% increase in unit sales for each of the four product lines. The CEO provides you the data.

ProductLine	LastYear	ThisYear
Computers	122322	132889
Phones	789112	998325
Printers	67891	32445
TVs	322112	401223

1. What data relationship are you exploring?
2. What visuals would you use to explore this relationship?
3. What insights can you collect from your exploration?

BE 7.8 (LO 2, 3) Managerial Accounting QB (Quality Breads) is a bakery in downtown Los Angeles selling artisanal breads. The estimated daily fixed costs are $4,000. The cost to make one loaf of bread is $3, and the selling price for each loaf is $5. As a result, the daily breakeven point for QBs, ignoring unsold bread, is 2,000 loaves of bread. Unsold bread, or spoilage, is disposed of at the end of the day. At the end of last week, QB generated the following report.





Discuss the data relationships explored in this report.

BE 7.9 **(LO 2, 3)** **Data** **Financial Accounting** CITREX is a biotech company in Miami, FL. One of the goals in their 2020 strategic plan was to improve their accounts receivable turnover ratio to 10. The table shows (net) sales and (average) accounts receivable information from their 2021–2025 financial statements.

Account	2021	2022	2023	2024	2025
Sales	$ 3,129,666	$ 3,518,891	$ 4,015,671	$ 5,913,556	$ 6,123,411
Accounts Receivable	$ 425,662	$ 436,771	$ 491,223	$ 549,901	$ 618,891

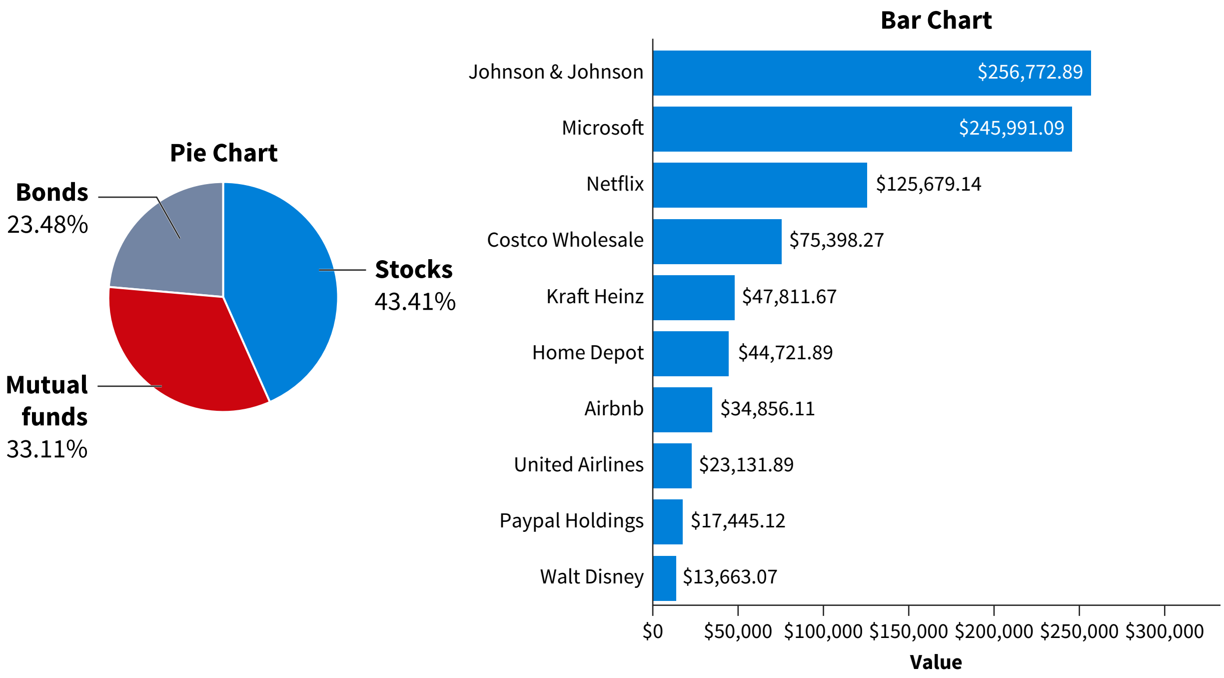
Explore CITREX's success in accomplishing this goal. Create two visualizations. Briefly describe what data relationships they represent and the insights you derived from each.

BE 7.10 **(LO 3)** **Data** **Financial Accounting** Poisson is a manufacturer of swimwear. They provide you the following financial summary for the last five years.

Account	2021	2022	2023	2024	2025
Revenues	$ 3,476,811	$ 3,337,714	$ 3,197,633	$ 3,159,912	$ 3,087,761
Gross Profit	$ 1, 890,556	$ 1,735,551	$ 1,640,941	$ 1,554,891	$ 1,438,664
Net Profit	$ 753,891	$ 827,714	$ 970,022	$ 1,000,033	$ 1,110,989

Explore the data. Create two visualizations. Briefly describe what data relationships they represent and the insights you derived from each.

BE 7.11 **(LO 2, 3)** Napigem is a Florida biotech company. They invest some of their assets in stocks, bonds, and mutual funds. The following dashboard helps them manage their investment portfolio.



1. What data relationships are explored in this dashboard?
2. Complete the exploration structure for this dashboard using the following field names.

Field Name	Description
InvestmentName	The name of a specific stock, mutual fund, or bond (e.g., Home Depot).
InvestmentType	Type of investment: bond, mutual fund, or stock.
TotalValue	The total portfolio value ($)

The same field name can be used more than once. (Some of the areas in the exploration structure will remain empty.)

Pie Chart

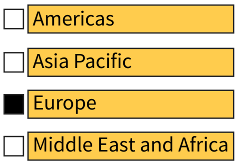
Filters	Columns
Rows	**Values**

Bar Chart

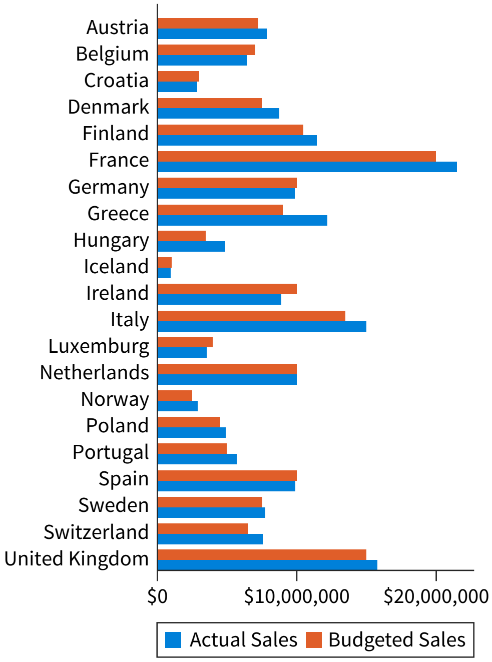
Filters	Columns
Rows	**Values**

BE 7.12 (LO 3) Sof*tee is company with headquarters in Dallas, TX, that sells baby care products worldwide. They created the following dashboard to make it easy for the C-Suite to monitor sales.

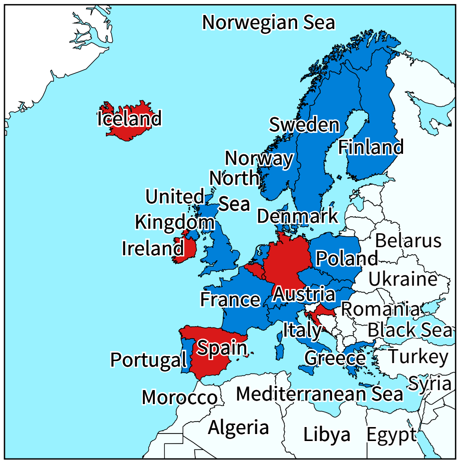
Slicer



Clustered Bar Chart



Map



1. What data relationships are explored in this dashboard?
2. Complete the exploration structure for this dashboard using the following field names.

Field Name	Description
ActualSales	Total actual sales.
BudgetedSales	Total budgeted sales in dollars.
Country	Name of the country.
Region	Name of the regio.
Variance	Total sales – total budgeted sales

Each field name can be used more than once. There will be some blank areas in the completed exploration structure.

Slicer

Filters	Columns
Rows	**Values**

Clustered Bar Chart

Filters	Columns
Rows	**Values**

Map

Filters	Columns
Rows	**Values**

Exercises

EX 7.1 (LO 1) Data Tax Accounting **Explore Tax Expenses** Instant Pest Control (IPC) is a medium-sized company with offices in Delaware, Maryland, New Jersey, and Pennsylvania. IPC's controller wants two tax-related questions answered.

1. For tax purposes, IPC is required to file Form 1099s for contractors and Form W-2s for employees. How many of both forms do they need to prepare for each state? Because IPC pays their tax accountants per prepared form, this information can help the controller estimate the fee to be paid.
2. IPC has established an internal policy to hire only in-state employees to simplify its tax filings. The state the office is located in and the employee's state of residency should be the same. Per state, to what extent are the offices compliant with this policy (as a percentage)?

EX 7.2 (LO 2) Data Managerial Accounting **Explore Salary Distribution** The Benford Group is a U.S. top-50 public accounting firm with more than 1,400 employees. Their managing partner, Tanya, provides you with a list of the salaries of all employees. What can you tell Tanya about the salary distribution at the Benford Group?

EX 7.3 (LO 2) Data Financial Accounting **Growth Analysis** Shrek, a mid-size bike manufacturer in Oxford, PA, targets a 5% growth in 2025 for all its product lines. As Shrek's controller, you have received the data set generated by the company's ERP system.

ProductLine	2024 UnitsSold	2025 UnitsSold
Adventure	2771	5741
City	8001	6799
Cyclocross	13998	14881
Family	23488	22476
Fitness	8112	11665
Mountain	7500	7430
Tandem	1089	722
Touring	3801	4022

1. Starting from the data you received, what are two key questions you would like to answer?
2. What data relationship is being explored?
3. What visualizations can be used for exploration purposes?
4. What insights are generated by your analysis?

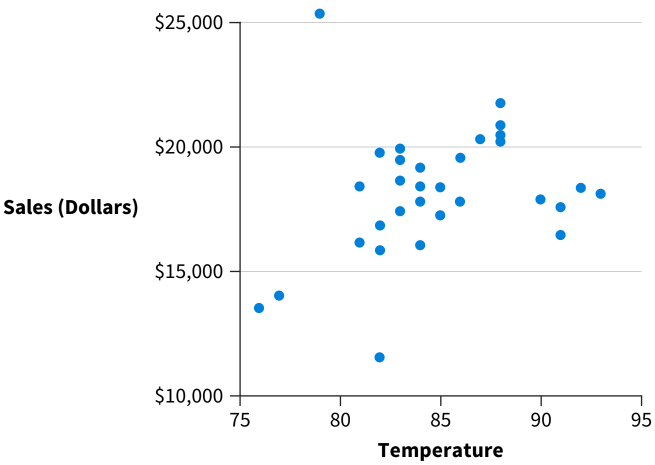
EX 7.4 (LO 2, 3) Data Managerial Accounting **Daily Sales Comparison** Meier is a chain of grocery stores. Their store in Big Rapids, MI gives you their 2025 February sales data for four products: deli, meat, chicken, and fish. A sample is shown next.

Day	Deli	Meat	Chicken	Fish
2/1/2025	$ 27,234	$ 87,665	$ 17,074	$ 5,992
2/2/2025	$ 34,889	$ 43,141	$ 38,827	$ 10,436
2/3/2025	$ 19,067	$ 61,675	$ 15,447	$ 12,024
2/4/2025	$ 17,812	$ 46,855	$ 32,029	$ 5,285
2/5/2025	$ 22,357	$ 42,584	$ 31,342	$ 9,361
2/6/2025	$ 11,525	$ 55,635	$ 42,466	$ 9,019
2/7/2025	$ 12,052	$ 38,495	$ 41,091	$ 24,889
2/8/2025	$ 28,554	$ 82,009	$ 24,739	$ 7,563
2/9/2025	$ 41,776	$ 45,106	$ 15,990	$ 12,569
2/10/2025	$ 16,200	$ 44,461	$ 25,780	$ 11,666

Create a visualization for each of the two questions they ask you to explore:

1. How do weekdays rank based on total February sales?
2. How does the sales share of each product compare across weekdays?

EX 7.5 (LO 2) Data Identify and Explore Data Relationships Ice Cow is a popular ice cream store in New York City's theater district. They created the following report for their July 2025 sales.



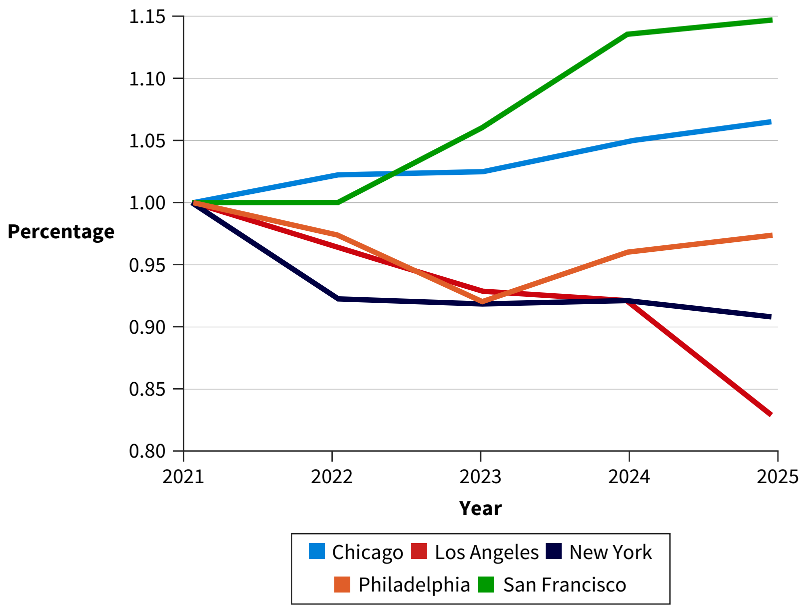
Date	Temperature	Sales
7/1/25	82	$ 16,820
7/2/25	81	$ 16,128
7/3/25	81	$ 18,412
7/4/25	79	$ 25,373
7/5/25	83	$ 17,396
7/6/25	85	$ 17,236
7/7/25	83	$ 18,628
7/8/25	84	$ 18,393
7/9/25	86	$ 17,773
7/10/25	88	$ 21,761
7/11/25	88	$ 20,411
7/12/25	91	$ 17,558
7/13/25	93	$ 18,104
7/14/25	82	$ 11,498
7/15/25	76	$ 13,497
7/16/25	77	$ 13,985
7/17/25	82	$ 15,825
7/18/25	84	$ 16,031
7/19/25	88	$ 20,248
7/20/25	92	$ 18,333
7/21/25	91	$ 16,442
7/22/25	90	$ 17,872
7/23/25	88	$ 20,923
7/24/25	82	$ 19,752
7/25/25	83	$ 19,919
7/26/25	84	$ 19,153
7/27/25	83	$ 19,494
7/28/25	85	$ 18,361
7/29/25	87	$ 20,311
7/30/25	86	$ 19,544
7/31/25	84	$ 17,790

1. What question is being explored?
2. What data relationship is being explored?
3. What insights does the data relationship generate?

EX 7.6 (LO 2, 3) Data Managerial Accounting **Identify and Explore Data Relationships** Gummy is a chain with candy trucks selling in five cities: Chicago, Los Angeles, New York, Philadelphia, and San Francisco. They provide you the following report.

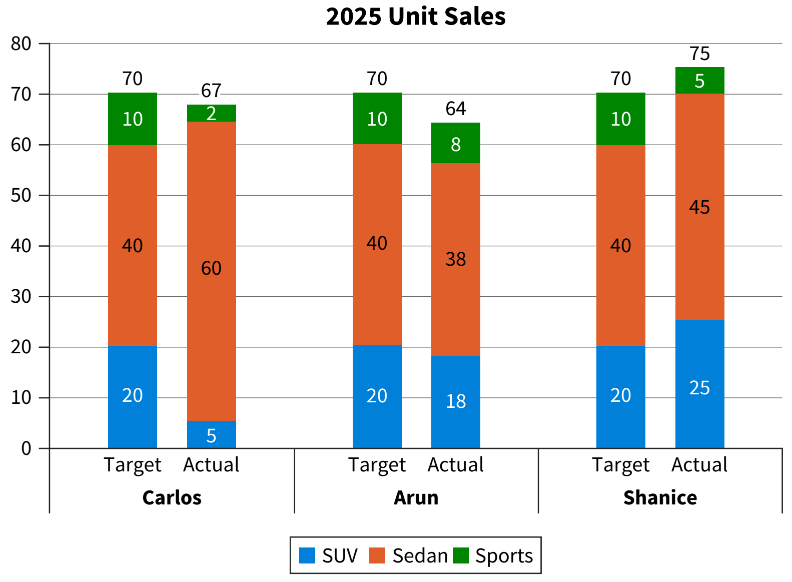
Number of Candy Trucks

Year	Chicago	Los Angeles	New York	Philadelphia	San Francisco
2021	1,240	1,712	1,350	750	920
2022	1,267	1,650	1,245	730	920
2023	1,270	1,590	1,240	690	975
2024	1,300	1,575	1,243	720	1,044
2025	1,320	1,420	1,225	730	1,055



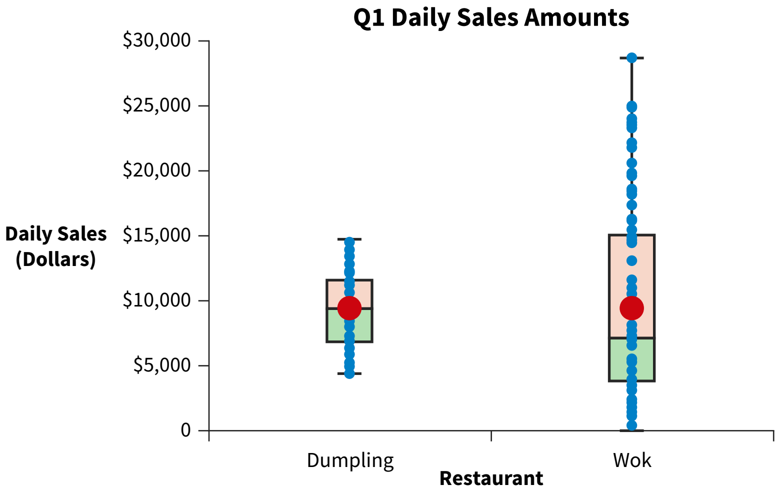
1. What question is being explored?
2. What data relationships are being explored?
3. What insights do the data relationships generate?

EX 7.7 (LO 2, 3) Data Managerial Accounting **Identify and Explore Data Relationships** Cook Cars is a small car dealership that employs three salespeople: Carlos, Arun, and Shanice. The dealership's owner asked you to prepare the following report.



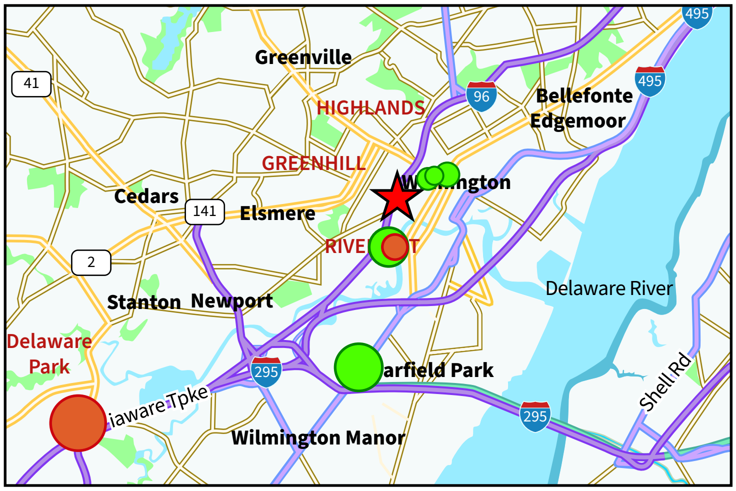
1. What questions are being explored?
2. What data relationships are being explored?
3. What insights do the data relationships generate?

EX 7.8 (LO 2) Data Identify and Explore Data Relationships Maggie Lee owns two take-out restaurants called Wok and Dumpling. Wok is located in downtown Denton, Texas. Dumpling is in the business district of the same city. Both restaurants serve a similar menu. At the end of the first quarter, Maggie receives a report she requested from her accountant.



1. What questions are being explored?
2. What data relationship is being explored?
3. What insights does the data relationship generate?

EX 7.9 (LO 2, 3) Identify and Explore Data Relationships The Santorini group, a public accounting firm in Amarillo, TX, has five employees who plan to attend a seminar in Wilmington, DE, next month. Their data analyst generates this report. The orange bubbles represent Hilton hotels, and green bubbles represent Marriott hotels. Bubble size indicates the rates charged: the smaller the bubble, the cheaper the nightly rate. The red star indicates the seminar location.



Chain	Hotel	Location	Rate/Night
Marriott	Courtyard Wilmington Downtown	1102 N West Street, Wilmington, Delaware, 19801	127
Hilton	DoubleTree by Hilton Hotel Wilmington	4727 Concord Pike, Wilmington, DE 19803	118
Marriott	Fairfield Inn & Suites Wilmington New Castle	2117 N Dupont Hwy, New Castle, DE 19720-6308	202
Hilton	Homewood Suites by Hilton Wilmington Downtown	820 Justison St, Wilmington, DE 19801	145
Hilton	Hotel Wilmington/Christiana	100 Continental Drive Newark, Delaware 19713-4301, USA	224
Marriott	Residence Inn Wilmington Downtown	1300 North Market Street, Wilmington, Delaware 19801	142
Marriott	Sheraton Suites Wilmington Downtown	422 Delaware Avenue, Wilmington, Delaware, 19801	136
Marriott	The Westin Wilmington	818 Shipyard Drive, Wilmington, Delaware 19801	175

1. What questions are being explored?
2. What data relationships are being explored?
3. What insights do the data relationships generate?

EX 7.10 (LO 2, 3) Data Identify and Explore Data Relationships Perez & Sons (PS) are the exclusive northeast distributors for South Track treadmills. There are four types of treadmills: M1750, M2450, M2950, and M3750. PS has four warehouses, one each in Boston, New York, Philadelphia, and Baltimore. To meet demand, they want 1,300 units of each type of treadmill in stock at any time, for a total of 5,200. They provide you with the data set.

Warehouse	Product	QOH
Boston	M1750	412
New York	M1750	288
Philadelphia	M1750	298
Baltimore	M1750	396
Boston	M2450	49
New York	M2450	658
Philadelphia	M2450	321
Baltimore	M2450	199
Boston	M2950	420
New York	M2950	519
Philadelphia	M2950	344
Baltimore	M2950	618
Boston	M3750	400
New York	M3750	400
Philadelphia	M3750	150
Baltimore	M3750	55

The three diagrams show possible exploration structures for this data set: A, B, and C.

(A)

Filters	Columns
	Product
Rows	**Values**
Warehouse	TotalQuantity OnHand

(B)

Filters	Columns
Product	
Rows	**Values**
Warehouse	TotalQuantity OnHand

(C)

Filters	Columns
TotalQuantity OnHand	
Rows	**Values**
Product	TotalQuantity OnHand

Do the following for each exploration structure:

1. Identity a data relationship that could be explored.
2. Determine questions that could be answered.
3. Create a visualization to support your analysis.
4. Explain insights that could be generated.

EX 7.11 (LO 2, 3) Data Financial Accounting Identify and Explore Data Relationships Bethany Beach, a small beach community, has three ice cream stores: Laureen's, Softer-Ice, and YuMeeS. The following data summarizes their revenue for the 2021 to 2025 period.

Bethany Beach Ice Cream Stores
Revenues
2021–2025

Store	2021	2022	2023	2024	2025
Laureen's	$ 283,776	$ 309,466	$ 340,011	$ 418,966	$ 498,007
Softer-Ice	$ 76,000	$ 123,000	$ 266,128	$ 314,000	$ 478,326
YuMeeS	$ 211,984	$ 192,000	$ 163,000	$ 149,000	$ 129,000

Generate at least three insights from the perspective of Laureen's. For each insight, discuss what data relationship you explored and what visual you used to generate it.

EX 7.12 Data Tax Accounting **Identify and Explore Data Relationships** The following table was taken from the IRS 2018 Statistics of Income report. It summarizes the Income Subject to Tax and Total Income Tax After Credits amounts on the federal returns filed by active corporations.

Size Total Assets	Income Subject To Tax	Total Income Tax After Credits
Zero Assets	$ 55,884,102	$ 11,594,836
$1 under $500,000	$ 7,829,008	$ 1,569,228
$500,000 under $1,000,000	$ 4,829,204	$ 956,103
$1,000,000 under $5,000,000	$ 17,223,201	$ 3,496,487
$5,000,000 under $10,000,000	$ 10,086,326	$ 2,015,713
$10,000,000 under $25,000,000	$ 19,143,340	$ 3,805,232
$25,000,000 under $50,000,000	$ 14,384,822	$ 2,811,835
$50,000,000 under $100,000,000	$ 16,854,446	$ 3,293,822
$100,000,000 under $250,000,000	$ 25,807,956	$ 4,772,611
$250,000,000 under $500,000,000	$ 26,192,384	$ 4,679,319
$500,000,000 under $2,500,000,000	$ 126,273,758	$ 20,835,589
$2,500,000,000 or more	$ 1,632,169,980	$ 184,855,832

Explore whether large companies with more assets have a higher tax rate.

1. What data relationship are you exploring?
2. Design a report with a visualization that enables you to explore this data relationship.
3. What insights can you collect from your exploration?

EX 7.13 **(LO 2, 3)** Data Tax Accounting **Exploring R&D Tax Credit Allocations** In 1997, the state of Pennsylvania enacted a research and development (R&D) tax credit to increase R&D activities within the commonwealth to enhance economic growth. The state's budget for the tax year 2020 was $55 million. The legislation required the state to set aside $11 million to support startups and small businesses (Small). The remaining $44 million was to be awarded to other businesses (Not-Small). The 2021 report summarizes the 2020 R&D credit amount requested and awarded.

PENNSYLVANIA RESEARCH & DEVELOPMENT TAX CREDIT - 2021 REPORT

2020 RECIPIENTS

			R&D CREDITS - DECEMBER 2020 ($M)		
BUSINESS TYPE	APPLICANTS	% OF APPLICANTS	TENTATIVE	ACTUAL	% OF ACTUAL
SMALL	470	42.2%	$18.1	$11.0	20.0%
NOT SMALL	643	57.8%	$189.7	$44.0	80.0%
TOTAL	**1,113**	**100.0%**	**$207.7**	**$55.0**	**100.0%**

Source: Adapted from THE RESEARCH & DEVELOPMENT TAX CREDIT REPORT to the PENNSYLVANIA GENERAL ASSEMBLY.

- The Tentative column represents the taxpayers' total requested credit amount on their R&D credit applications.
- The Actual column represents the total amount awarded to those submitting applications.
- The actual amounts determine the proportional amount of the budgets allocated.

Create visualizations to answer the following three questions:

1. What percentage of the amount was requested by each business type?
2. How much of the budget was allocated to each business type, Small and Not Small?
3. What percentage of the money requested did each group receive?

EX 7.14 (LO 3) Data Perform Pareto Analysis Nicole, manager of the Rainbow hotel, provides you this summary table regarding customer complaints received in January 2025.

Type	Source	Number
Dirty Rooms	Hotel Website	4
Dirty Rooms	Calls	22
Elevator	Calls	2
Poor Wifi	Hotel Website	4
Poor Wifi	ratemyhotel.com	2
Poor Wifi	Calls	7
Rodents	Calls	2
Room Temperature	Calls	2
Unexpected Fees	Letter	14
Unexpected Fees	Hotel Website	60
Unexpected Fees	ratemyhotel.com	16
Unexpected Fees	Calls	90
Unfriendly Staff	Letter	2
Unfriendly Staff	Hotel Website	105
Unfriendly Staff	ratemyhotel.com	44
Unfriendly Staff	Calls	45

1. Use the data file to create a Pareto chart.
2. Discuss the insights it generates.

EX 7.15 Data Explore with an Interactive Report Jumpers is a grocery store in Indianapolis with locations in NIndy (North Indy) and SIndy (South Indy). They compiled a five-year report with revenues generated for each of their seven product categories (meat, drinks, vegetables and fruits, snacks, freezer, diary, other).

1. Create an interactive report based on the following exploration structure.

Filters	Columns
Store	
Rows	**Values**

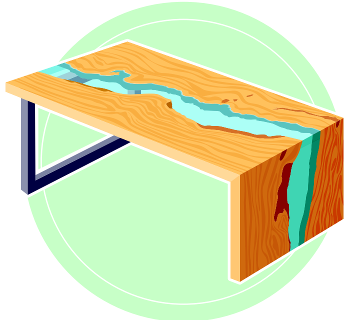
Filters	Columns
Store ProductName	Store
Rows	**Values**
Year	TotalAmount

Filters	Columns
ProductName	
Rows	**Values**

Filters	Columns
	ProductName
Rows	**Values**
Year	TotalAmount

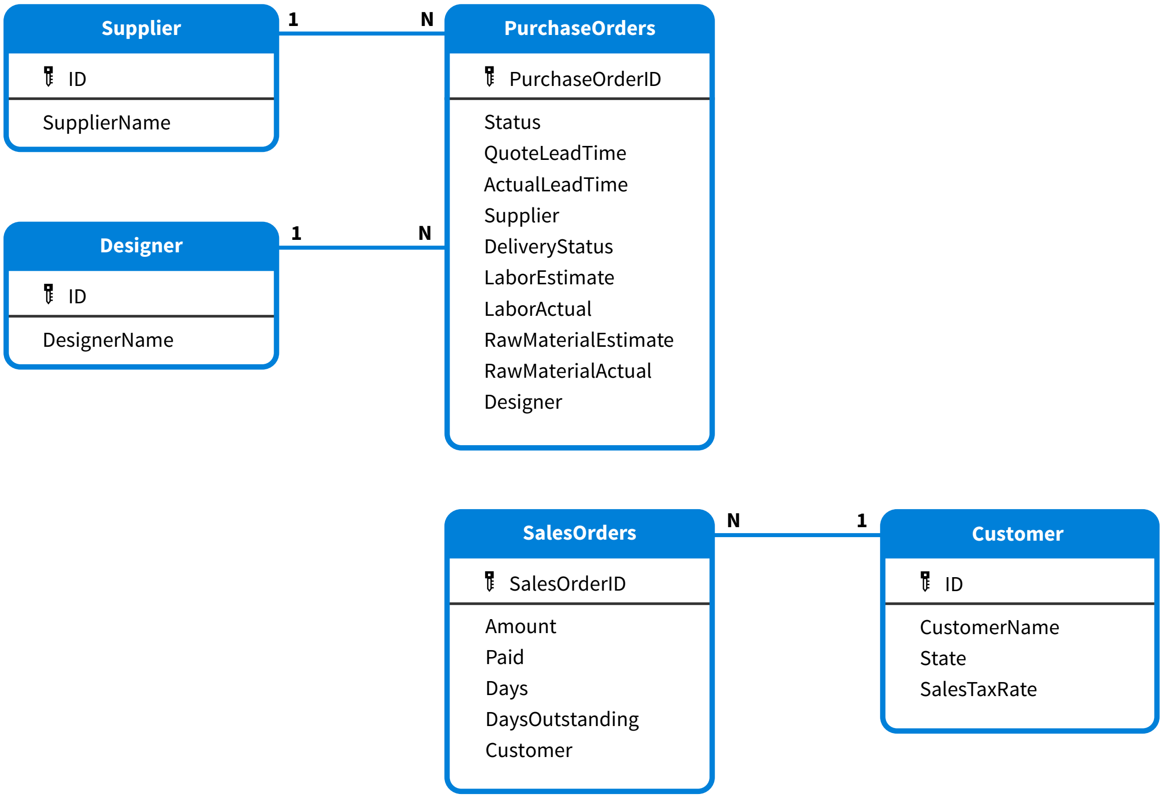
2. Describe the relationships that are part of your dashboard.
3. Identify questions that can be explored with your dashboard.

Professional Application Case: NoTable



NoTable is a furniture store in Halfway, Oregon, that manufactures custom-designed wooden tables of exceptional quality. Most of their customers are collectors, including several movie stars. Their business model is simple: a customer contacts them, they work with the customer on a design, build the table, and deliver it to the customer's residence. NoTable just hired you as an intern for their accounting department. They hope your data analytics skills can help them with some of the decisions they face. They provide you with an Excel file that contains several worksheets with data generated from their ERP system. The underlying data model is shown here, followed by a data dictionary.

Data Model



Data Dictionary

PurchaseOrders

Name	Description
PurchaseOrderId	Uniquely identifies a purchase order.
Status	A purchase order's status: Canceled, Delivered, Open, or Returned.
QuotedLeadTime	The number of days the vendor committed to deliver the goods.
ActualLeadTime	The number of days it took the vendor to deliver the goods.
Supplier	ID of the vendor from whom the goods are purchased.
DeliveryStatus	An order is canceled, delivered, open, or returned. For orders that have been delivered, it is specified whether they were early, ontime, late, or disruptive. Disruptive implies that the production process was disrupted.
LaborEstimate	The estimated dollar amount for labor for the production order.
LaborActual	The actual dollar amount used for labor for the production order.

(Continued)

Name	Description
RawMaterialEstimate	The estimated dollar amount for raw materials for the production order.
RawMaterialActual	The actual dollar amount for raw materials for the production order.
Designer	ID of the designer who designed the production order (table).

Supplier

Name	Description
ID	Uniquely identifies a vendor.
Name	A vendor's name.

SalesOrders

Name	Description
SalesOrderId	Uniquely identifies a sales order.
Amount	The amount the customer owes NoTable.
Paid	Has the order been paid in full: Yes or No.
Days	For orders that are fully paid, the number of days it took to pay them.
DaysOutstanding	For orders that are not fully paid yet, the number of days the receivable has been outstanding.
Customer	ID of the customer to whom the table was sold.

Customer

Name	Description
ID	Uniquely identifies a customer.
Name	A customer's name.
State	The state in which a customer lives.
SalesTaxRate	The sales tax rate for the state in which the customer lives.

Designer

Name	Description
ID	Uniquely identifies a designer.
Name	A designer's name.

PAC 7.1 Auditing: Explore Vendor Reliability

Data **Auditing** Part of NoTable's success rests on its use of premium wood such as Bubinga and Dalbergia. One of its main challenges is to find suppliers for such wood. They currently work with eleven different suppliers, and selection is exclusively based on price: the order goes to the supplier with the lowest price. You point out that supplier performance, particularly on-time delivery, should be considered as well. Risks related to late deliveries include lost sales, downtime, or not meeting the delivery schedule. Late deliveries can dramatically impact the bottom line. You decide to use the data to support your statement. You start by developing an algorithm to create a field called DeliveryStatus that contains one of the following values: Early, Ontime, Late, Disruptive, Canceled, Returned. You already added that column to the NoTable file.

1. Using the values in the DeliveryStatus field, explore the delivery risks NoTable is facing.
2. Assess the risk of each supplier, and assign them one the following labels: Safe, Risky, High-Risk.
3. To mitigate supplier risk, how would you suggest tiering NoTable's suppliers?

PAC 7.2 Managerial Accounting: Manage Variances

Data **Managerial Accounting** Another key part of NoTable's success is its designers, highly-paid artists who design the tables with and for customers. They currently have five designers, and one is assigned to each purchase order. They meet extensively with each of the customers and develop comprehensive specifications. Production steps and required raw materials are specified with the highest level of detail. Accuracy of cost estimates are of utmost importance for NoTable given that they determine a table's price.

You receive data generated by NoTable's ERP system that show the estimated and actual labor and material cost in dollars for each production order (table). You can find these data in the PurchaseOrders worksheet. You are asked to analyze how accurate the design specifications are both overall and among designers.

1. Determine and analyze the overall variances.
2. Compare variances among designers.

PAC 7.3 Financial Accounting: Manage Accounts Receivable

Data **Financial Accounting** NoTable's management would like to include additional information about credit risk in their financial report. They ask you to explore accounts receivable and provide some insights. They provide you with sales orders and payment data generated by their ERP system. You can find these data in the SalesOrders and Customer worksheets. The following is some additional information regarding NoTable's credit policy:

- All payments are due within 30 days.
- Early payments: There is a 3% discount for all orders paid in full within 2 weeks (15 days).
- On time payments: There is no discount or no penalty for sales orders paid on time.
- Late payments: There is a 3% penalty for all sales orders with a late payment.

1. Explore whether NoTable's credit policy works. For orders that already have been paid, compare early, on time, and late payments.
2. For the orders that have not been fully paid yet, explore the aging of the outstanding balances. What insights can you collect?
3. Analyze accounts receivable by customer. Explore whether there is an issue with accounts receivable concentration.

PAC 7.4 Tax Accounting: Analyze Sales Tax

Data **Tax Accounting** NoTable delivers items to customer residencies. You notice that no sales tax was collected for any of the sales.

1. Determine the total sales tax amount that NoTable should have collected.
2. Determine how much sales tax should have been collected and remitted to the different states.

Le Grind Continuing Case: Investigate Gross Profit Using Exploration Patterns

Data Visit Wiley's online learning platform for the case background, additional questions, data, and more details about the continuing case.